

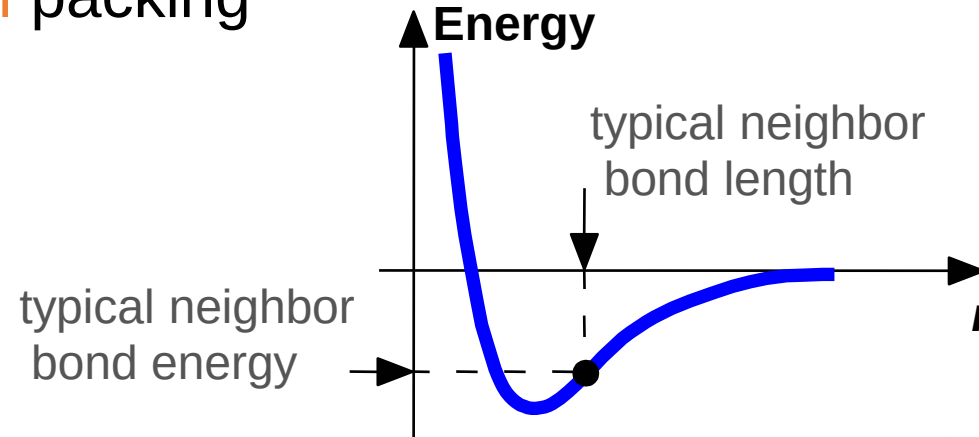
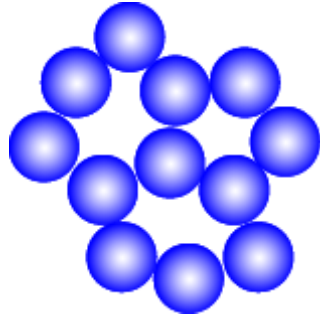
Chapter 3: The Structure of Crystalline Solids

ISSUES TO ADDRESS...

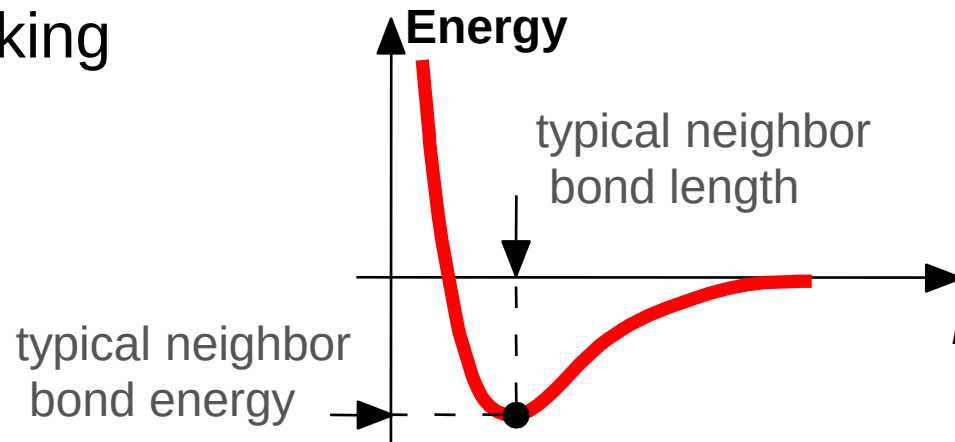
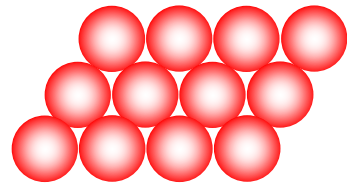
- How do atoms assemble into solid structures?
- How does the density of a material depend on its structure?
- When do material properties vary with the sample (i.e., part) orientation?

Energy and Packing

- Non dense, **random** packing



- Dense, **ordered** packing

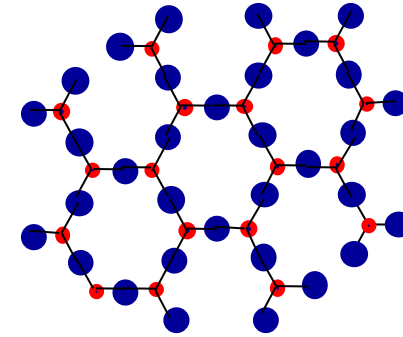


Dense, ordered packed structures tend to have lower energies.

Materials and Packing

Crystalline materials...

- atoms pack in periodic, 3D arrays
- typical of:
 - metals
 - many ceramics
 - some polymers



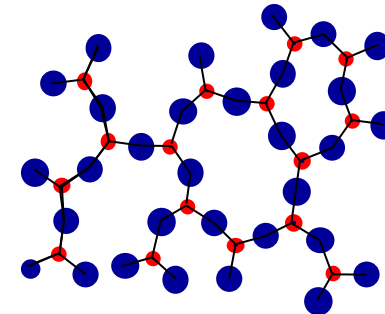
crystalline SiO₂

Adapted from Fig. 3.23(a),
Callister & Rethwisch 8e.

Noncrystalline materials...

- atoms have no periodic packing
- occurs for:
 - complex structures
 - rapid cooling

• Si • Oxygen



noncrystalline SiO₂

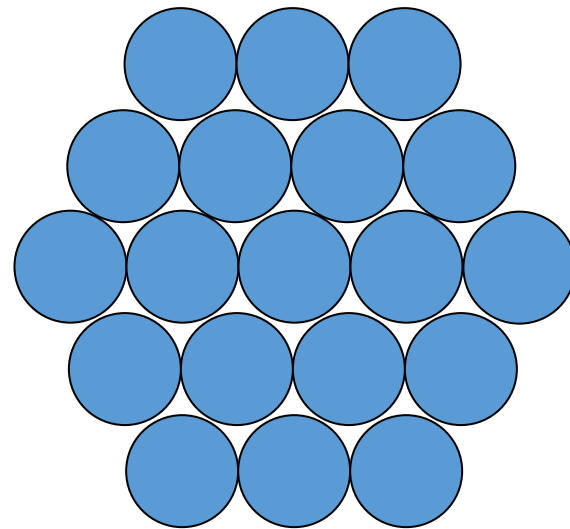
Adapted from Fig. 3.23(b),
Callister & Rethwisch 8e.

"Amorphous" = Noncrystalline

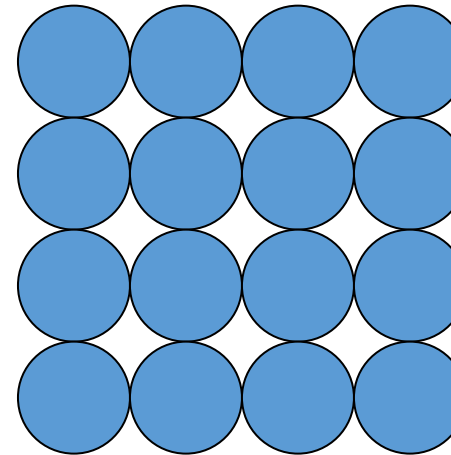
Metallic Crystal Structures

- How can we stack metal atoms to minimize empty space?

2-dimensions



vs.



Now stack these 2-D layers to make 3-D structures

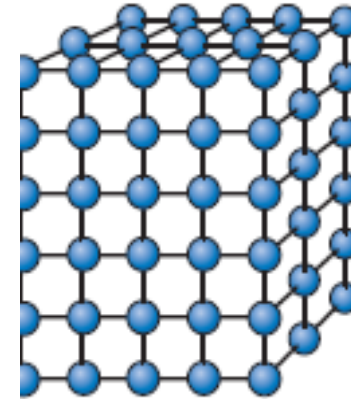
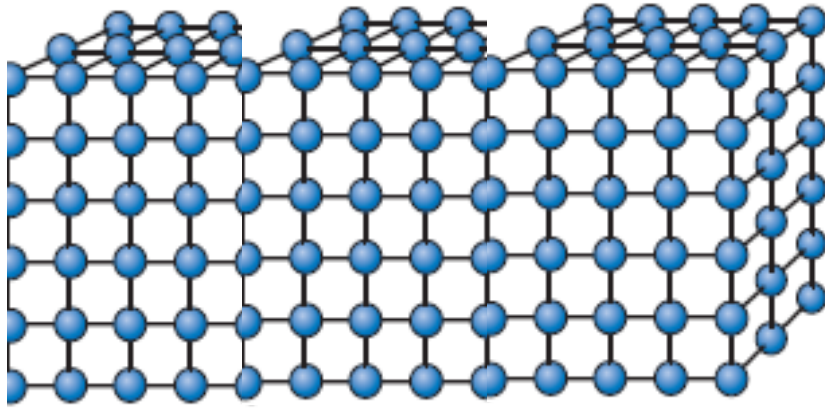
Metallic Crystal Structures

- Tend to be densely packed.
- Reasons for dense packing:
 - Typically, only one element is present, so all atomic radii are the same.
 - Metallic bonding is not directional.
 - Nearest neighbor distances tend to be small in order to lower bond energy.
 - Electron cloud shields cores from each other
- Have the simplest crystal structures.

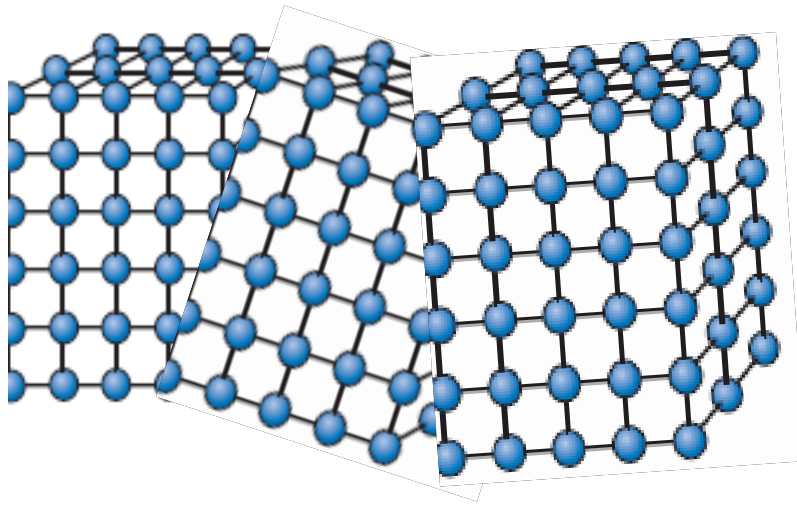
We will examine three such structures...

- No order: Monoatomic gases
- Short-Range Order (SRO): If order extends only to **nearest neighboring particles** e.g., in water molecules (hydrogen bonding), polymers, glasses.
- Long-Range Order (LRO): Materials with **a crystal structure** fall in the LRO category. Crystalline materials may consist of either:
 - A single crystal
 - Many misaligned crystals, called grains – these materials are called polycrystalline.

Long-Range Order



LR order Crystal



Long-Range Order

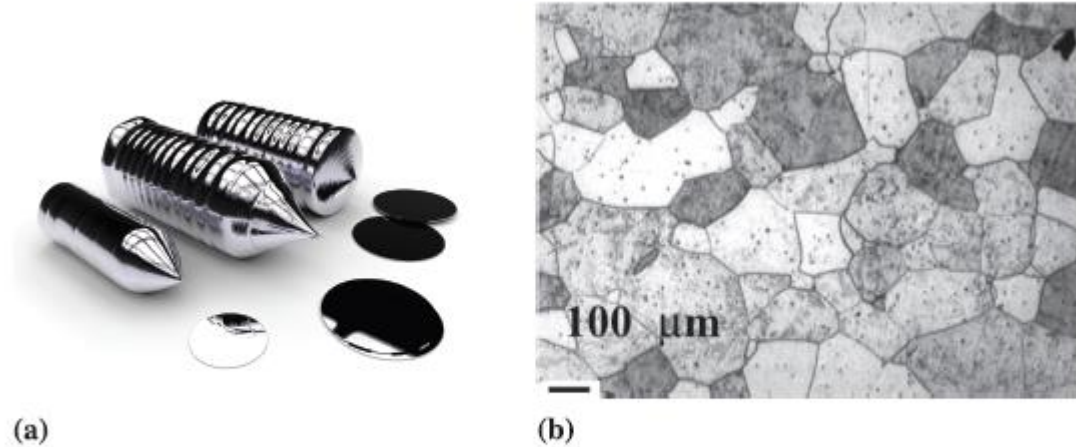


Figure 3-2 (a) Photograph of a silicon single crystal. (*Petr SobolevliStock/Thinkstock*)
(b) Micrograph of a polycrystalline stainless steel showing grains and grain boundaries.
(*Micrograph courtesy of Dr. A.J. Deardo, Dr. M. Hua and Dr. J. Garcia*)

Many engineering materials are polycrystalline,
e.g., steels.

Amorphous Materials

Any solid exhibiting **only SRO** of atoms/ions is an amorphous material.

In general, amorphous materials form when the **process kinetics prevent the material from reaching its LRO equilibrium state.**

Examples:

- Glasses typically form in polymer & ceramic systems.
- Metallic glasses, possessing interesting properties, may be formed by rapidly cooling metals/alloys to prevent crystallization.

Long-Range Order:

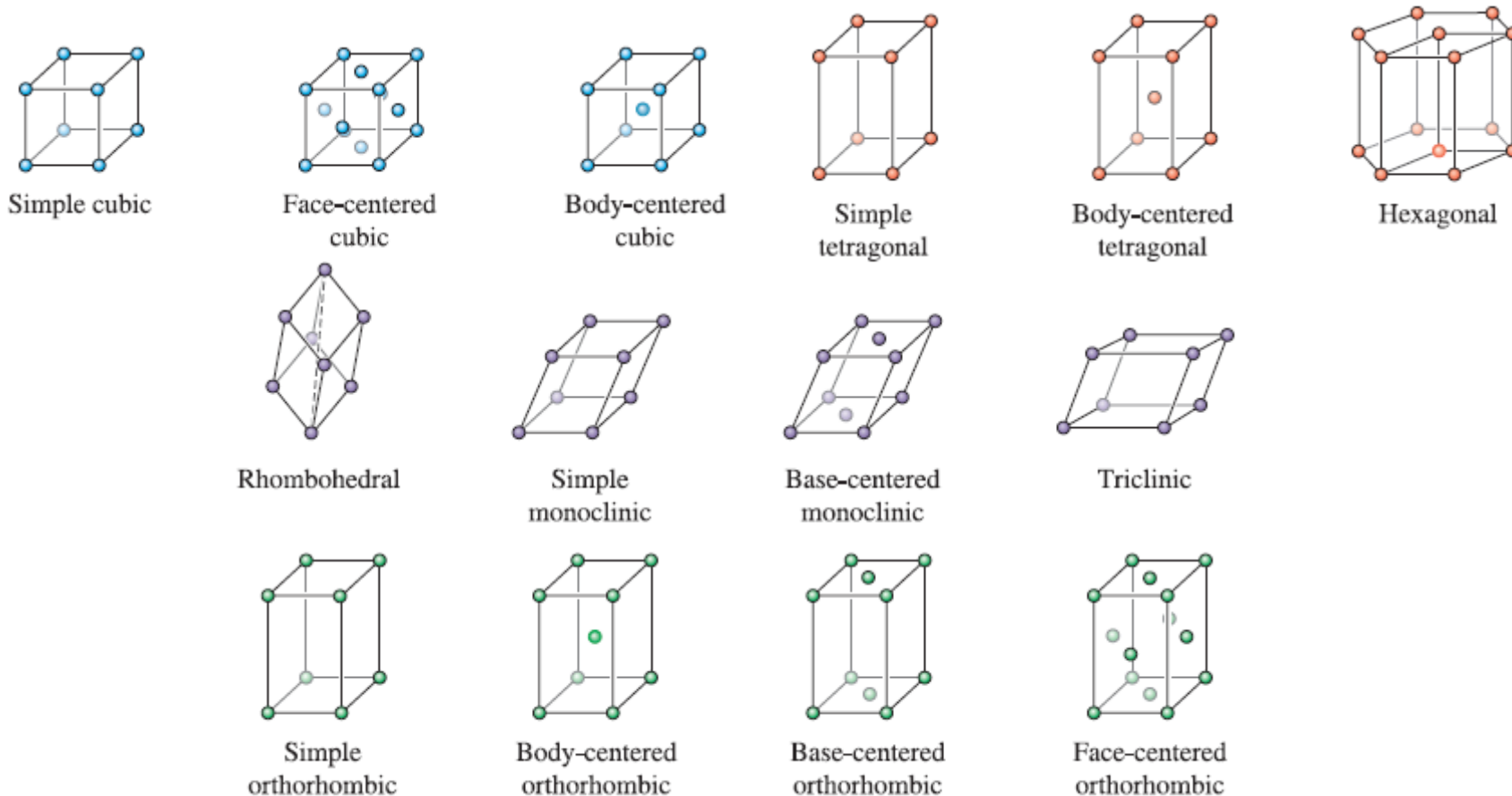
Crystal Structure (Lattice, Basis) & Unit Cells

- To describe the 3-D spatial arrangement of atoms/ions in a solid, we use the concept of crystal structure, lattice and basis.
- Crystal structure = **lattice** + **basis**
- **Lattice**: is a collection of points arranged in a periodic pattern, such that each lattice point's surroundings are identical (**Symmetry**)
- **Basis** is a group of 1 or more atoms arranged in a particular way with respect to each other and associated with each lattice point.

Lattice:

Lattice, Basis, Unit Cells & Crystal Structure

There are 14 unique ways to arrange points in 3D space. They are called the Bravais lattices.



Lattice, Basis, Unit Cells & Crystal Structure

Table 3-1 Characteristics of the seven crystal systems

Structure	Axes	Angles between Axes	Volume of the Unit Cell
Cubic	$a = b = c$	All angles equal 90° .	a^3
Tetragonal	$a = b \neq c$	All angles equal 90° .	a^2c
Orthorhombic	$a \neq b \neq c$	All angles equal 90° .	abc
Hexagonal	$a = b \neq c$	Two angles equal 90° . The angle between a and b equals 120° .	$0.866a^2c$
Rhombohedral or trigonal	$a = b = c$	All angles are equal and none equals 90° .	$a^3\sqrt{1 - 3\cos^2\alpha + 2\cos^3\alpha}$
Monoclinic	$a \neq b \neq c$	Two angles equal 90° . One angle (β) is not equal to 90° .	$abc \sin \beta$
Triclinic	$a \neq b \neq c$	All angles are different and none equals 90° .	$abc\sqrt{1 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma + 2\cos\alpha\cos\beta\cos\gamma}$

Unit Cells & Crystal Structure

close-packed directions: Directions within the unit cell in which the atoms are in continuous contact are called

We can calculate a lattice parameter a_0 by relating the dimensions of the unit cell to the size of the atoms, using close-packed directions.

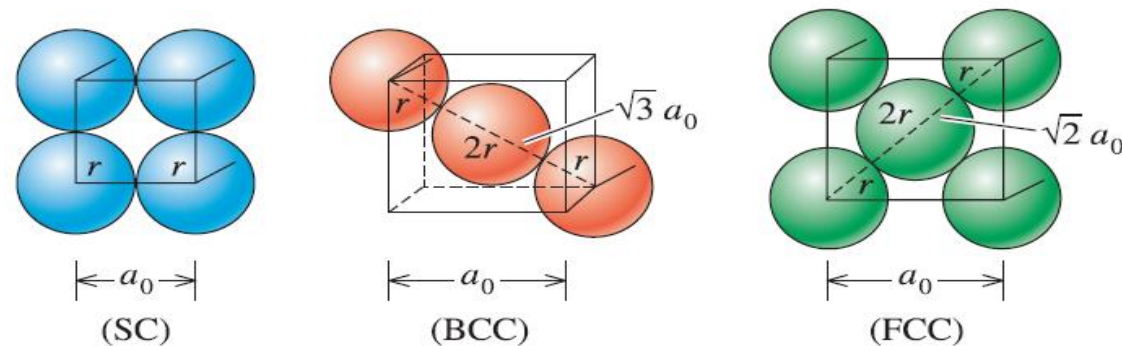
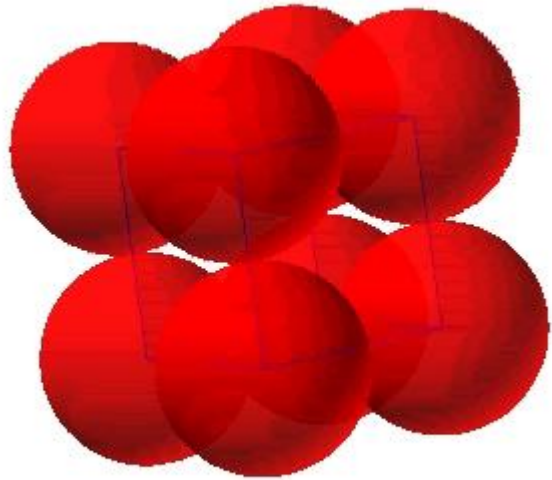


Figure 3-9 The relationships between the atomic radius and the lattice parameter in cubic systems (for Example 3-3).

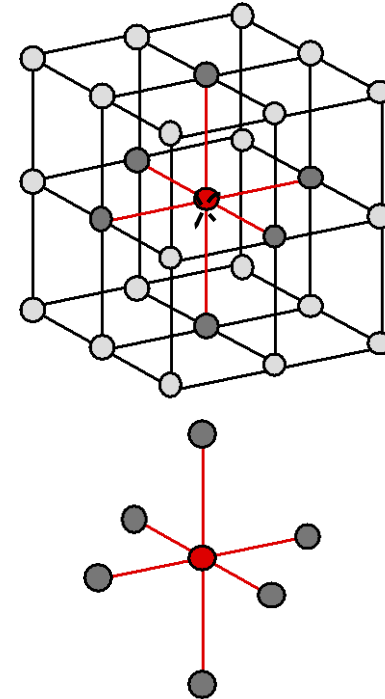
Simple Cubic Structure (SC)

- Rare due to low packing density (only Po has this structure)
- **Close-packed directions** are cube edges.



Click once on image to start animation
(Courtesy P.M. Anderson)

- **Coordination #** = 6
(# nearest neighbors)

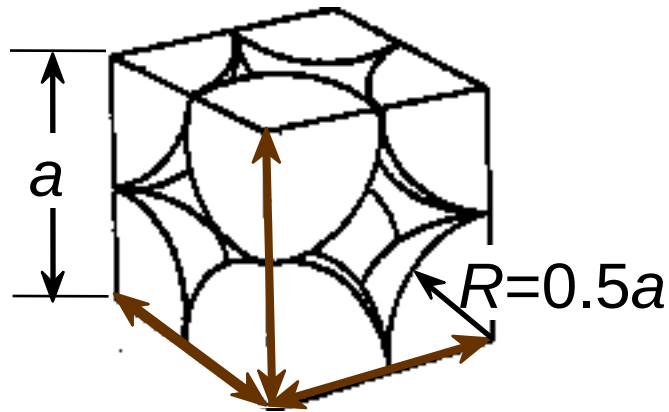


Atomic Packing Factor (APF)

$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

contains $8 \times 1/8 =$

1 atom/unit cell

Adapted from Fig. 3.24,
Callister & Rethwisch 8e.

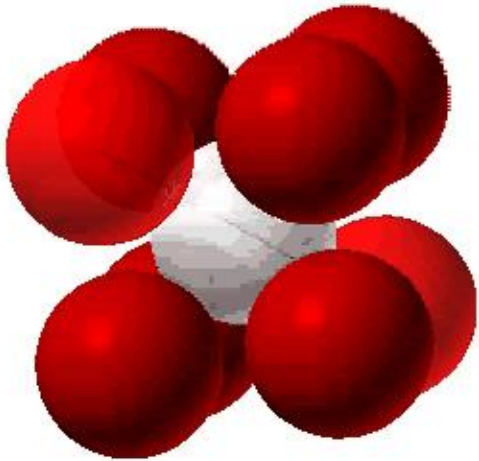
$$APF = \frac{\overbrace{1}^{\text{atoms}} \overbrace{\frac{4}{3} \pi (0.5a)^3}^{\text{volume atom}}}{\underbrace{a^3}_{\text{volume unit cell}}}$$

Body Centered Cubic Structure (BCC)

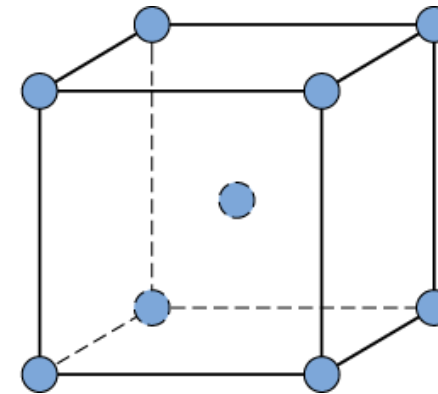
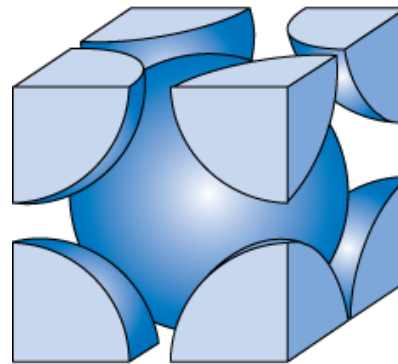
- Atoms touch each other along cube diagonals.
--Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.

ex: Cr, W, Fe (α), Tantalum, Molybdenum

- Coordination # = 8



Click once on image to start animation
(Courtesy P.M. Anderson)

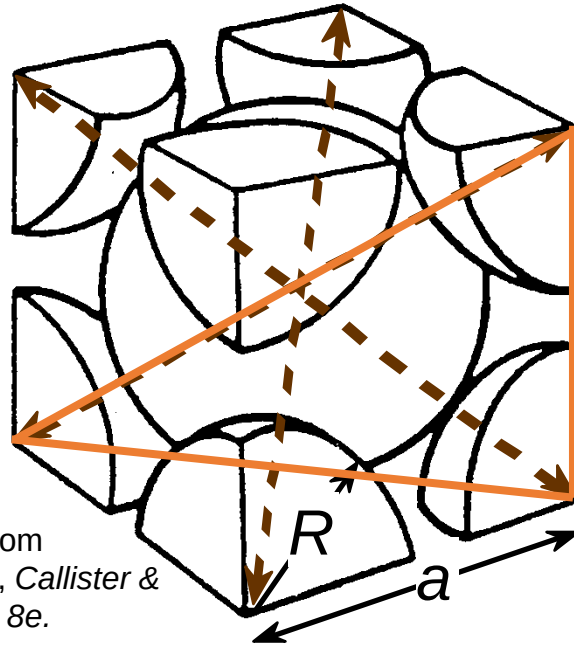


Adapted from Fig. 3.2,
Callister & Rethwisch 8e.

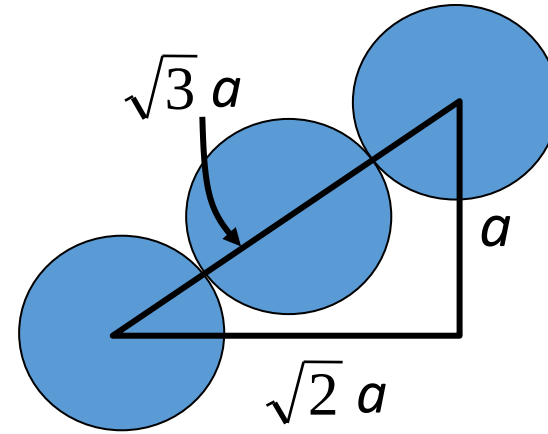
2 atoms/unit cell: 1 center + 8 corners $\times$ 1/8

Atomic Packing Factor: BCC

- APF for a body-centered cubic structure = 0.68



Adapted from
Fig. 3.2(a), Callister &
Rethwisch 8e.



Close-packed directions:
length = $4R = \sqrt{3}a$

$$\text{APF} = \frac{\text{atoms unit cell} \times \text{volume atom}}{\text{volume unit cell}}$$

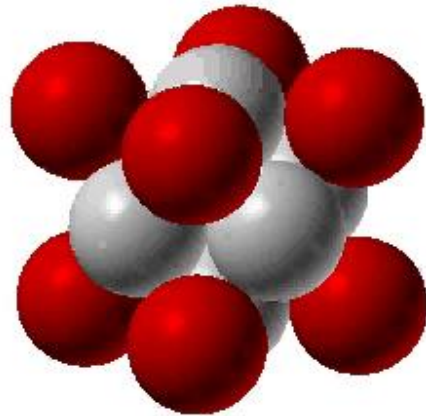
The diagram shows the calculation of the Atomic Packing Factor (APF) for BCC. The numerator consists of two parts: the number of atoms per unit cell (2) and the volume of a single atom ($\frac{4}{3} \pi (\sqrt{3}a/4)^3$). The denominator is the volume of the unit cell (a^3). Arrows point from the descriptive text to the corresponding terms in the equation.

Face Centered Cubic Structure (FCC)

- Atoms touch each other along face diagonals.
 - Note: All atoms are identical; the face-centered atoms are shaded differently only for ease of viewing.

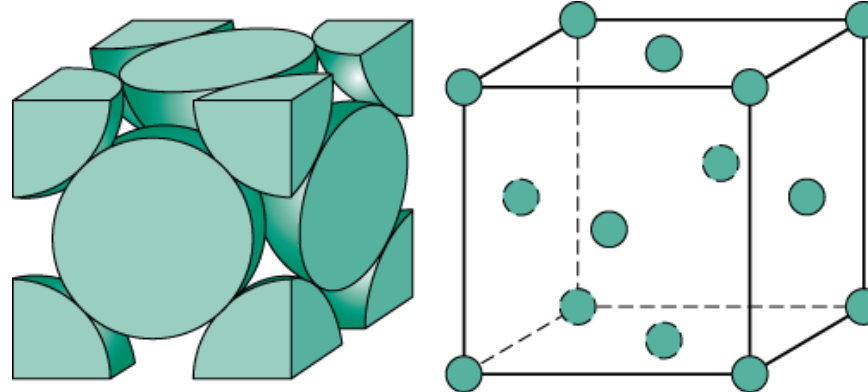
ex: Al, Cu, Au, Pb, Ni, Pt, Ag

- Coordination # = 12



Click once on image to start animation

(Courtesy P.M. Anderson)

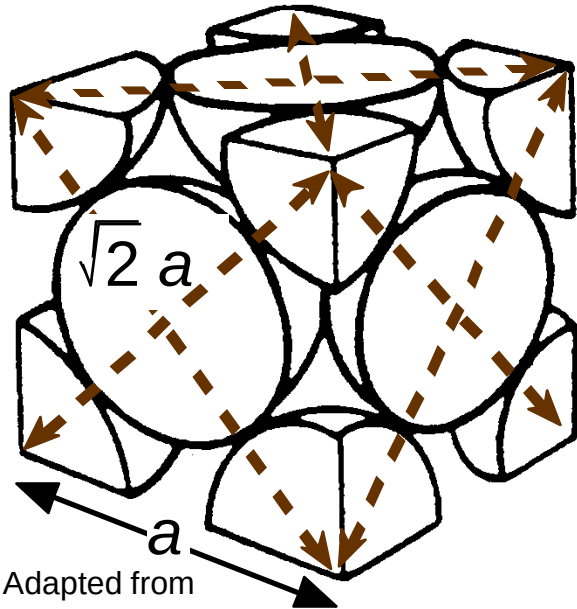


Adapted from Fig. 3.1, *Callister & Rethwisch 8e*.

4 atoms/unit cell: $6 \text{ face} \times \frac{1}{2} + 8 \text{ corners} \times \frac{1}{8}$

Atomic Packing Factor: FCC

- APF for a face-centered cubic structure = 0.74
maximum achievable APF



Adapted from
Fig. 3.1(a),
Callister &
Rethwisch 8e.

Close-packed directions:
length = $4R = \sqrt{2} a$

Unit cell contains:
 $6 \times 1/2 + 8 \times 1/8$
= 4 atoms/unit cell

$$\text{APF} = \frac{\text{atoms/unit cell} \times \text{volume/atom}}{\text{volume/unit cell}}$$

The equation is visually represented with colored boxes and arrows:

- The numerator consists of a green box containing '4' (labeled 'atoms/unit cell' with a green arrow) and an orange box containing $\frac{4}{3} \pi (\sqrt{2}a/4)^3$ (labeled 'volume/atom' with a brown arrow).
- The denominator is a blue box containing a^3 (labeled 'volume/unit cell' with a blue arrow).

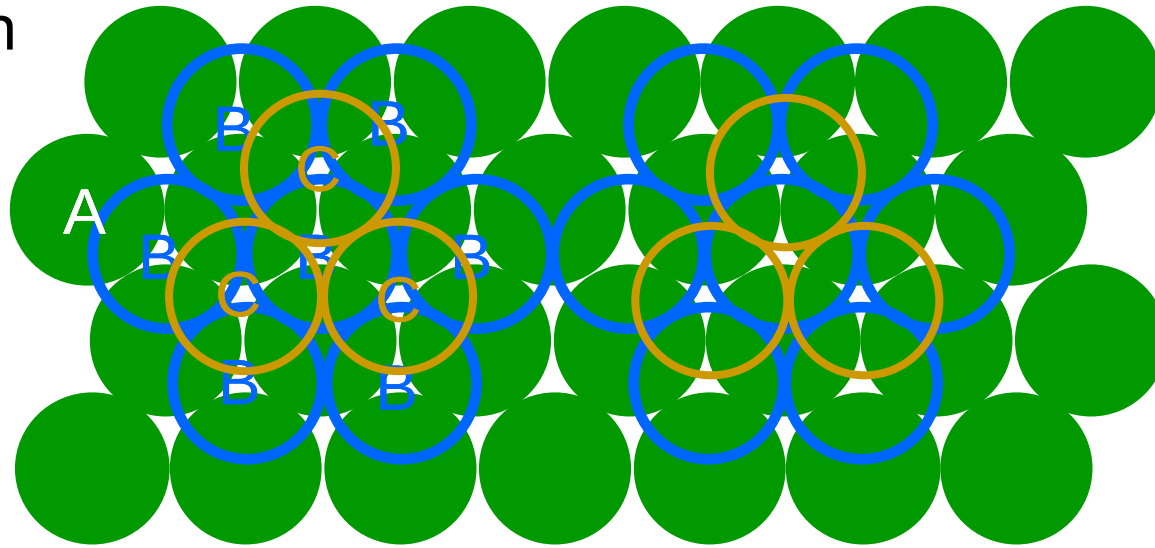
FCC Stacking Sequence

- ABCABC... Stacking Sequence
- 2D Projection

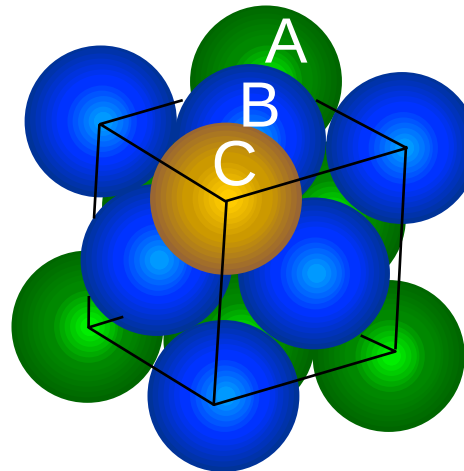
A sites

B sites

C sites

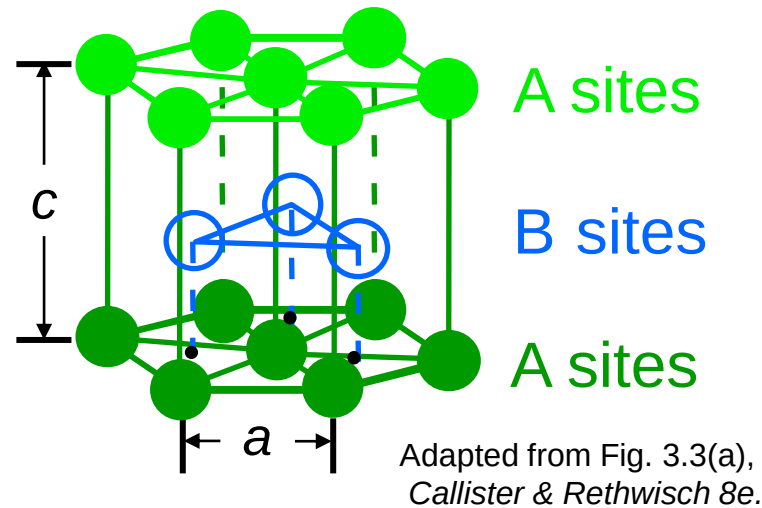


- FCC Unit Cell



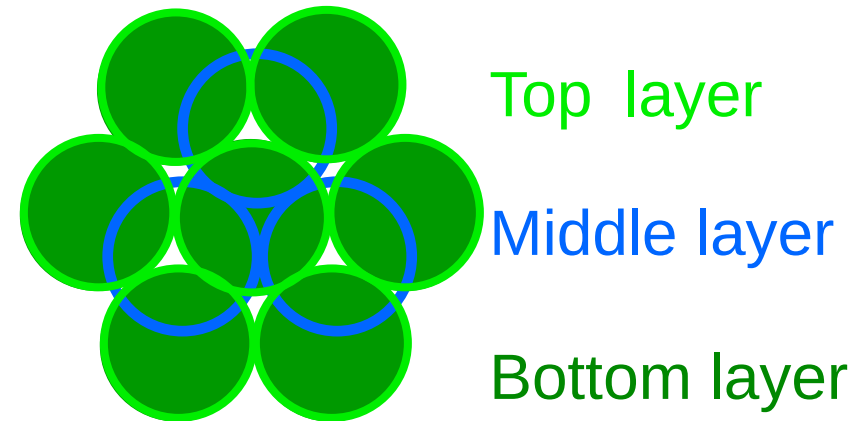
Hexagonal Close-Packed Structure (HCP)

- ABAB... Stacking Sequence
- 3D Projection



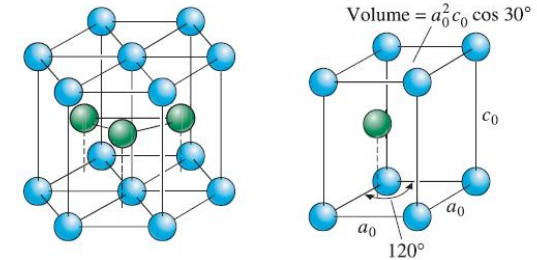
- Coordination # = 12
- APF = 0.74
- $c/a = 1.633$

- 2D Projection



6 atoms/unit cell

ex: Cd, Mg, Ti, Zn



Lattice, Basis, Unit Cells & Crystal Structure

Table 3-2 Crystal structure characteristics of some metals at room temperature

Structure	a_0 versus r	Atoms per Cell	Coordination Number	Packing Factor	Examples
Simple cubic (SC)	$a_0 = 2r$	1	6	0.52	Polonium (Po), α -Mo
Body-centered cubic (BCC)	$a_0 = 4r / \sqrt{3}$	2	8	0.68	Fe, W, Mo, Nb, Ta, K, Na, V, Cr
Face-centered cubic (FCC)	$a_0 = 4r / \sqrt{2}$	4	12	0.74	Cu, Au, Pt, Ag, Pb, Ni
Hexagonal close-packed (HCP)	$a_0 = 2r$ $c_0 \approx 1.633a_0$	2	12	0.74	Ti, Mg, Zn, Be, Co, Zr, Cd

Theoretical Density, ρ

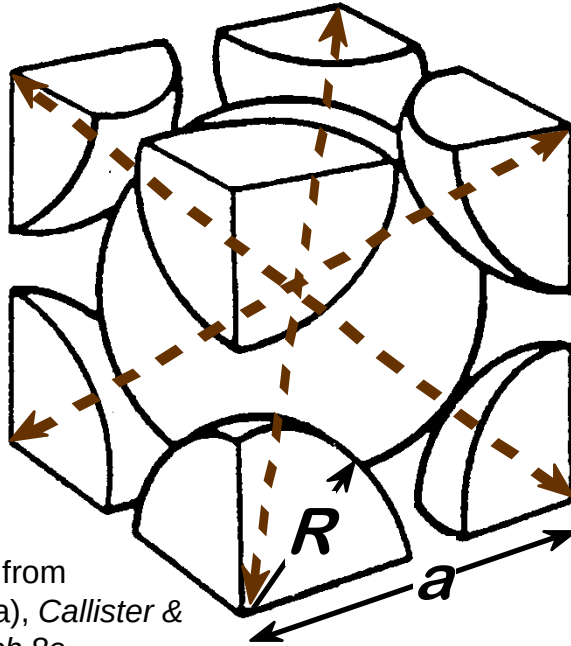
$$\text{Density} = \rho = \frac{\text{Mass of Atoms in Unit Cell}}{\text{Total Volume of Unit Cell}}$$

$$\rho = \frac{n A}{V_C N_A}$$

where

- n = number of atoms/unit cell
- A = atomic weight
- V_C = Volume of unit cell = a^3 for cubic
- N_A = Avogadro's number
= 6.022×10^{23} atoms/mol

Theoretical Density, ρ



Adapted from
Fig. 3.2(a), Callister &
Rethwisch 8e.

- Ex: Cr (BCC)

$$A = 52.00 \text{ g/mol}$$

$$R = 0.125 \text{ nm}$$

$$n = 2 \text{ atoms/unit cell}$$

$$a = 4R/\sqrt{3} = 0.2887 \text{ nm}$$

$$\rho = \frac{\frac{\text{atoms}}{\text{unit cell}} \times \frac{\text{g}}{\text{mol}}}{\frac{\text{volume}}{\text{unit cell}} \times \frac{\text{atoms}}{\text{mol}}}$$

$\frac{\text{atoms}}{\text{unit cell}} = 2$
 $\frac{\text{g}}{\text{mol}} = 52.00$
 $\frac{\text{volume}}{\text{unit cell}} = a^3$
 $\frac{\text{atoms}}{\text{mol}} = 6.022 \times 10^{23}$

$\rho_{\text{theoretical}}$	$= 7.18 \text{ g/cm}^3$
ρ_{actual}	$= 7.19 \text{ g/cm}^3$

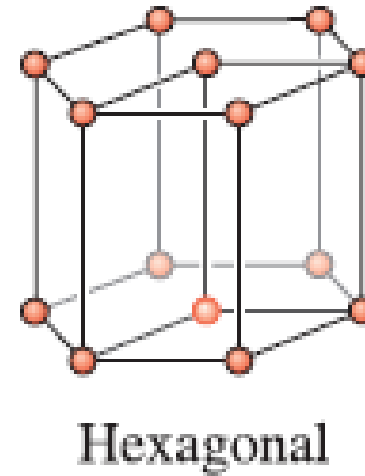
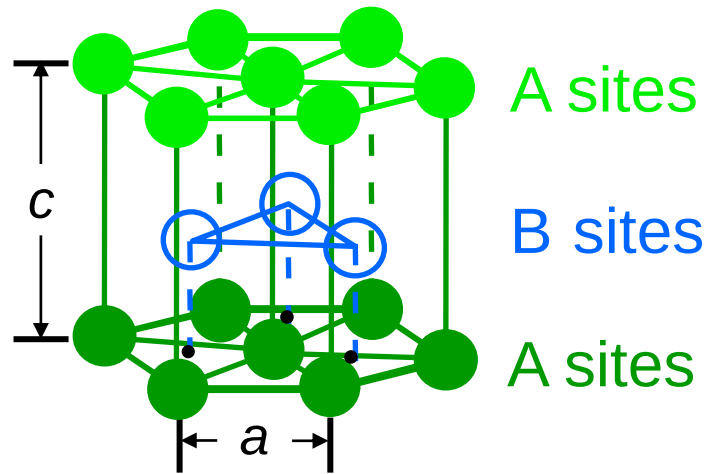
Theoretical Density, ρ

Ex: Cu (FCC)

$$A = 63.50 \text{ g/mol}$$

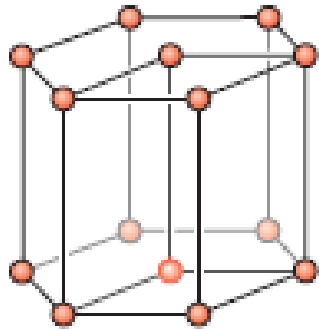
$$R = 0.128 \text{ nm}$$

Bravais lattice of HCP structure

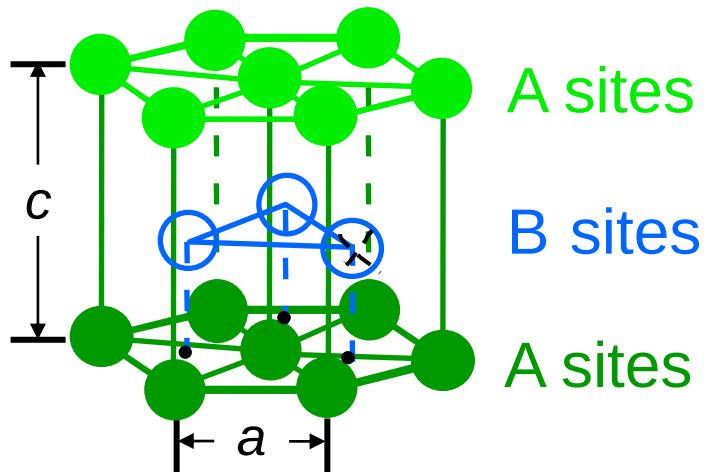


Atom positions

Basis



Hexagonal

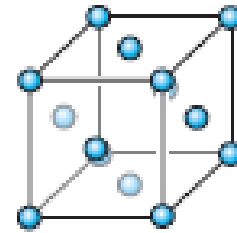


A sites

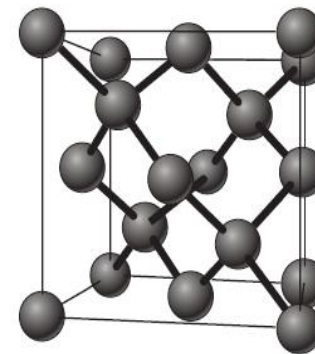
B sites

A sites

Atom positions
Lattice types
Lattice + Basis



Face-centered
cubic



Diamond

Densities of Material Classes

In general

$$\rho_{\text{metals}} > \rho_{\text{ceramics}} > \rho_{\text{polymers}}$$

Why?

Metals have...

- close-packing
(metallic bonding)
- often large atomic masses

Ceramics have...

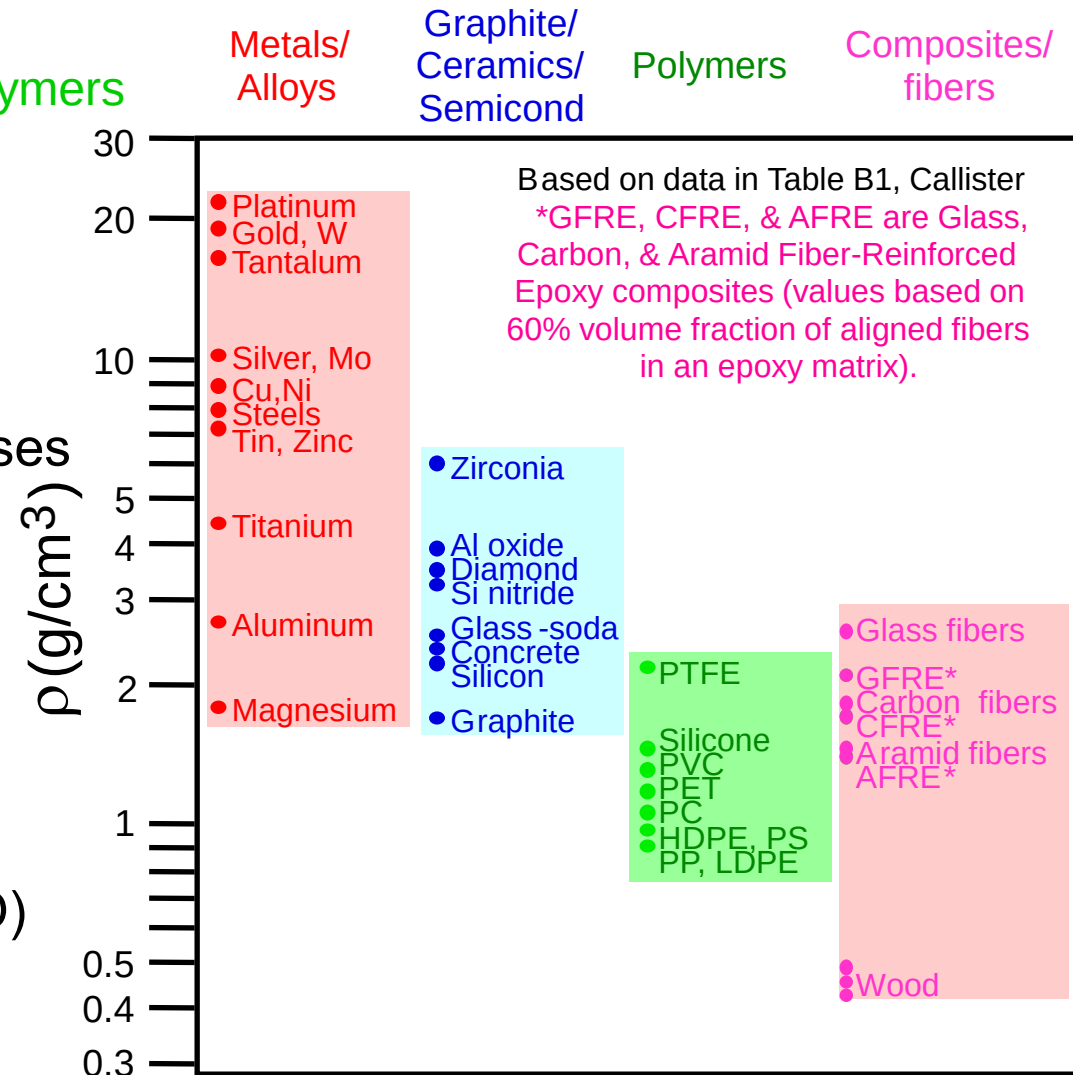
- less dense packing
- often lighter elements

Polymers have...

- low packing density
(often amorphous)
- lighter elements (C,H,O)

Composites have...

- intermediate values



Data from Table B.1, Callister & Rethwisch, 8e.

Crystals as Building Blocks

- *Some* engineering applications require single crystals:
 - diamond single
 - turbine blades

crystals for abrasives



(Courtesy Martin Deakins, GE Superabrasives, Worthington, OH. Used with permission.)

Fig. 8.33(c), *Callister & Rethwisch 8e.* (Fig. 8.33(c) courtesy of Pratt and Whitney).



- Properties of crystalline materials often related to crystal structure.
 - Ex: Quartz fractures more easily along some crystal planes than others.



(Courtesy P.M. Anderson)

Polycrystals

- Most engineering materials are **polycrystals**.



Anisotropic

Adapted from Fig. K,
color inset pages of
Callister 5e.
(Fig. K is courtesy of
Paul E. Danielson,
Teledyne Wah Chang
Albany)

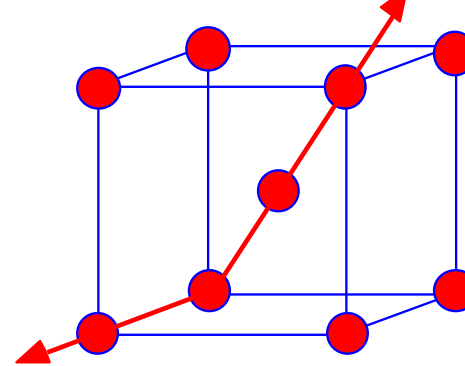
Isotropic

- Nb-Hf-W plate with an electron beam weld.
- Each "grain" is a single crystal.
- If grains are randomly oriented,
overall component properties are not directional.
- Grain sizes typically range from 1 nm to 2 cm
(i.e., from a few to millions of atomic layers).

Single vs Polycrystals

- Single Crystals
 - Properties vary with direction: **anisotropic**.
 - Example: the modulus of elasticity (E) in BCC iron:

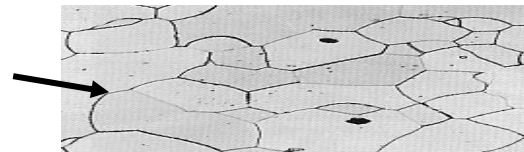
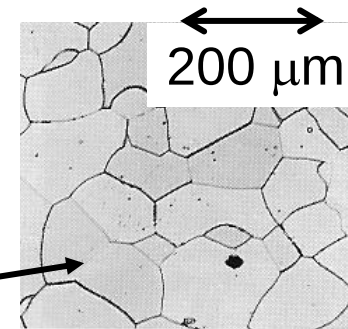
E (diagonal) = 273 GPa



E (edge) = 125 GPa

Data from Table 3.3, *Callister & Rethwisch 8e*. (Source of data is R.W. Hertzberg, *Deformation and Fracture Mechanics of Engineering Materials*, 3rd ed., John Wiley and Sons, 1989.)

- Polycrystals
 - Properties may/may not vary with direction.
 - If grains are randomly oriented: **isotropic**. ($E_{\text{poly iron}} = 210 \text{ GPa}$)
 - If grains are **textured**, anisotropic.



Adapted from Fig. 4.14(b), *Callister & Rethwisch 8e*. (Fig. 4.14(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC [now the National Institute of Standards and Technology, Gaithersburg, MD].)

Polymorphism

- Two or more distinct crystal structures for the same material (allotropy/polymorphism)

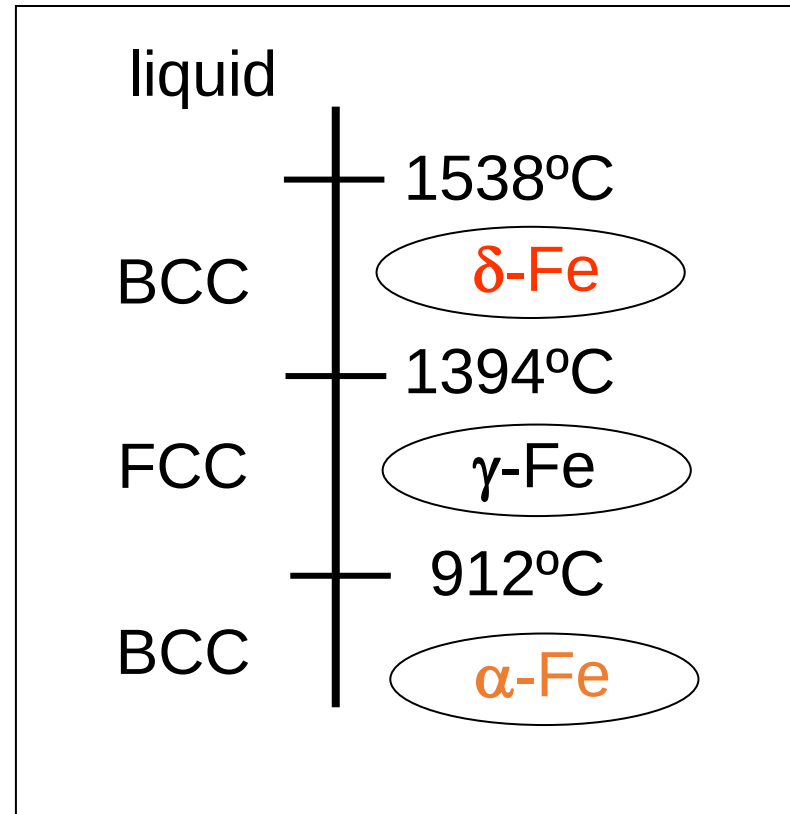
titanium

α , β -Ti

carbon

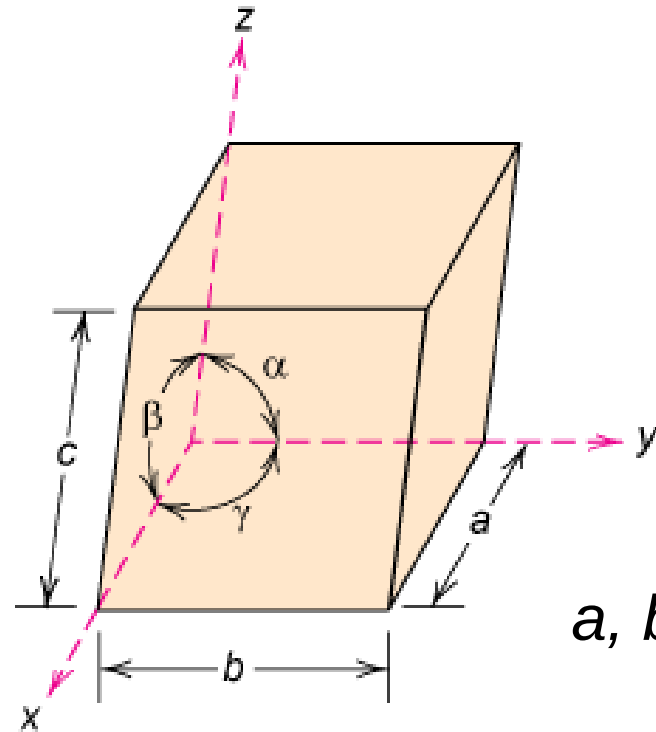
diamond, graphite

iron system



Crystal Systems

Unit cell: smallest repetitive volume which contains the complete lattice pattern of a crystal.



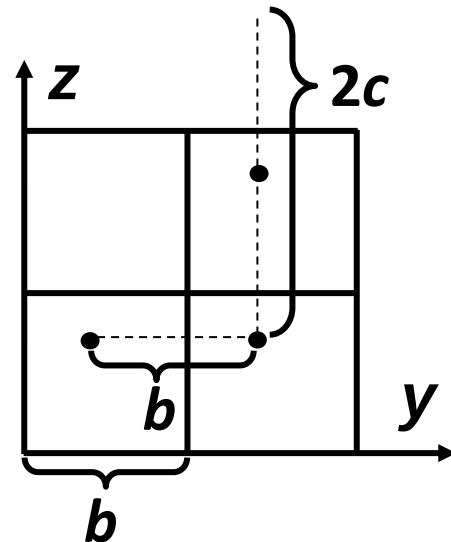
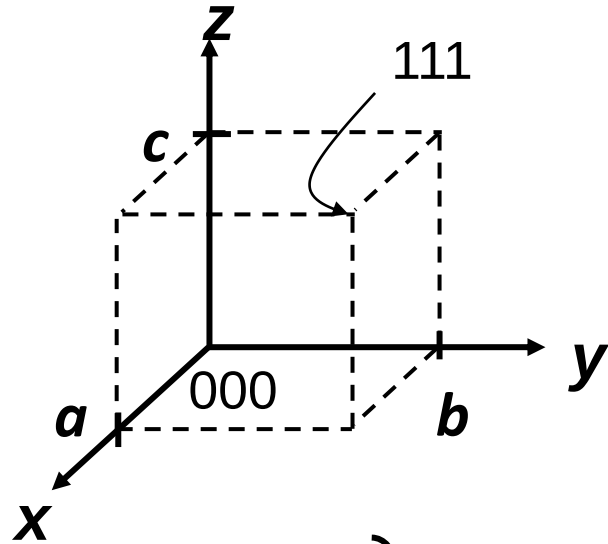
7 crystal systems

14 crystal lattices

a , b , and c are the lattice constants

Fig. 3.4, Callister & Rethwisch 8e.

Point Coordinates



Point coordinates for unit cell center are

$$a/2, b/2, c/2 \quad \frac{1}{2} \frac{1}{2} \frac{1}{2}$$

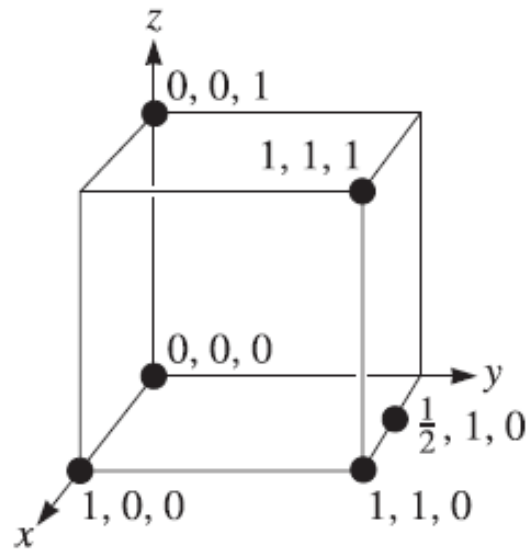
Point coordinates for unit cell corner are 111

Translation: integer multiple of lattice constants $\rightarrow$ identical position in another unit cell

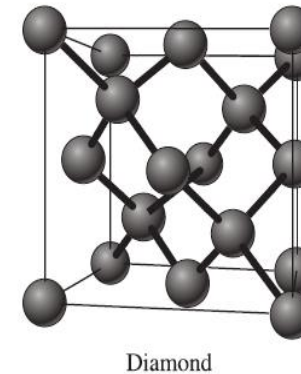
Coordinates of atoms in the Unit Cell

We can locate atom positions in the unit cell, by a right-handed coordinate system.

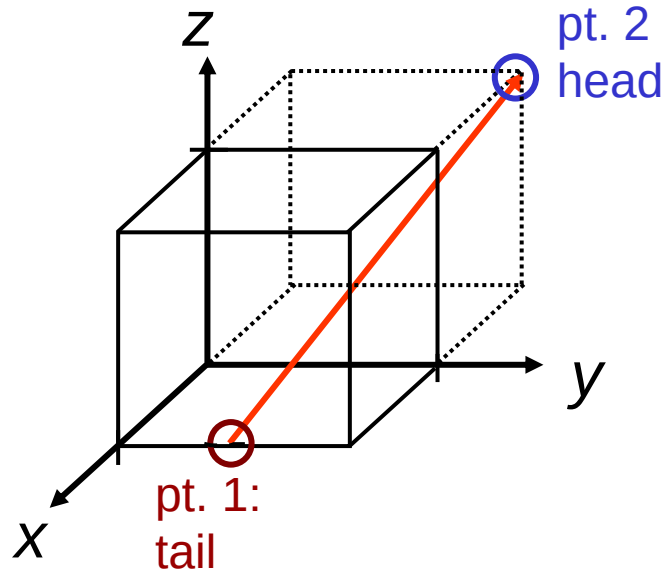
Basis



Atom positions
Lattice types
Lattice + Basis



Crystallographic Directions



Example 2:

pt. 1 $x_1 = a, y_1 = b/2, z_1 = 0$

pt. 2 $x_2 = -a, y_2 = b, z_2 = c$

$$\frac{-a - a}{a} \quad \frac{b - b/2}{b} \quad \frac{c - 0}{c}$$

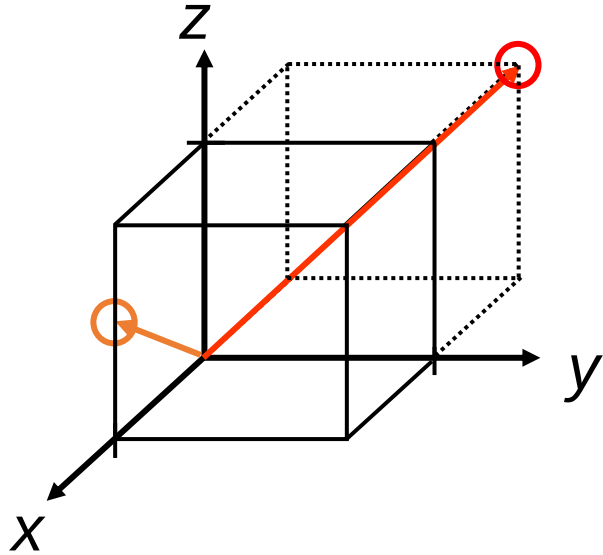
$$\Rightarrow -2, 1/2, 1$$

Multiplying by 2 to eliminate the fraction

$$-4, 1, 2 \Rightarrow [\bar{4}12] \quad \text{where the overbar represents a negative index}$$

families of directions $\langle uvw \rangle$

Crystallographic Directions



Algorithm

1. Vector repositioned (if necessary) to pass through origin.
2. Read off projections in terms of unit cell dimensions a , b , and c
3. Adjust to smallest integer values
4. Enclose in square brackets, no commas

$[uvw]$

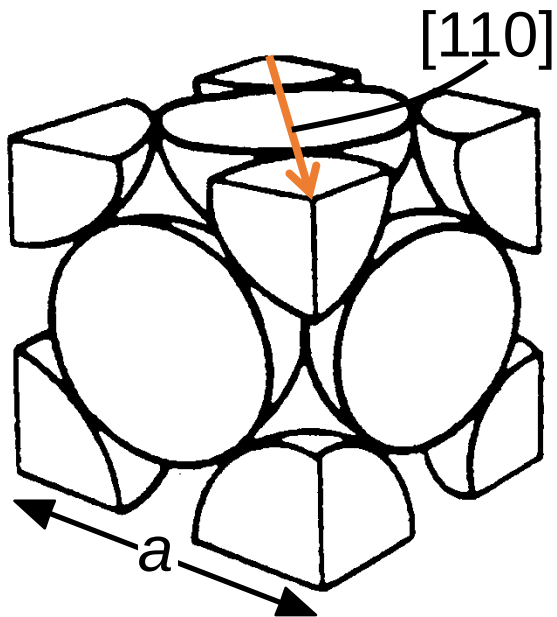
ex: $1, 0, \frac{1}{2} \Rightarrow 2, 0, 1 \Rightarrow [201]$

$-1, 1, 1 \Rightarrow [\bar{1}11]$ where overbar represents a negative index

families of directions $\langle uvw \rangle$

Linear Density

- Linear Density of Atoms $\equiv$ LD =
$$\frac{\text{Number of atoms}}{\text{Unit length of direction vector}}$$



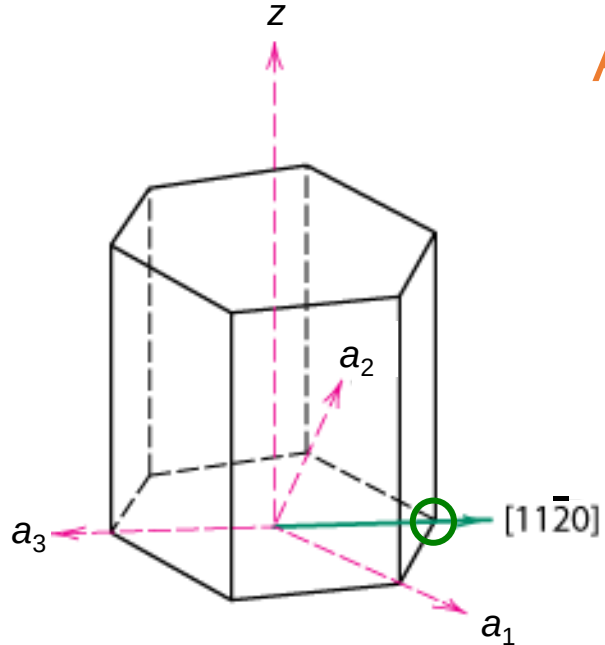
Adapted from
Fig. 3.1(a),
Callister &
Rethwisch 8e.

ex: linear density of Al in [110]
direction

$$a = 0.405 \text{ nm}$$

$$\text{LD} = \frac{\overset{\text{\# atoms}}{2}}{\underset{\text{length}}{\sqrt{2}a}} = 3.5 \text{ nm}^{-1}$$

HCP Crystallographic Directions



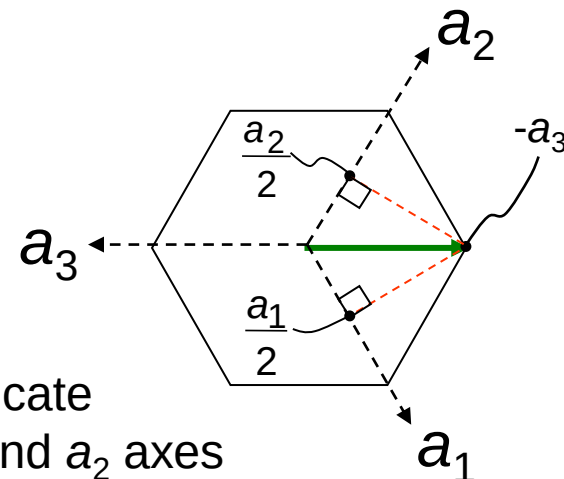
Adapted from Fig. 3.8(a),
Callister & Rethwisch 8e.

Algorithm

1. Vector repositioned (if necessary) to pass through origin.
2. Read off projections in terms of unit cell dimensions a_1 , a_2 , a_3 , or c
3. Adjust to smallest integer values
4. Enclose in square brackets, no commas

$[uvw]$

ex: $\frac{1}{2}, \frac{1}{2}, -1, 0 \Rightarrow [11\bar{2}0]$



dashed red lines indicate
projections onto a_1 and a_2 axes

HCP Crystallographic Directions

- Hexagonal Crystals
 - 4 parameter Miller-Bravais lattice coordinates are related to the direction indices (i.e., $u'v'w'$) as follows.

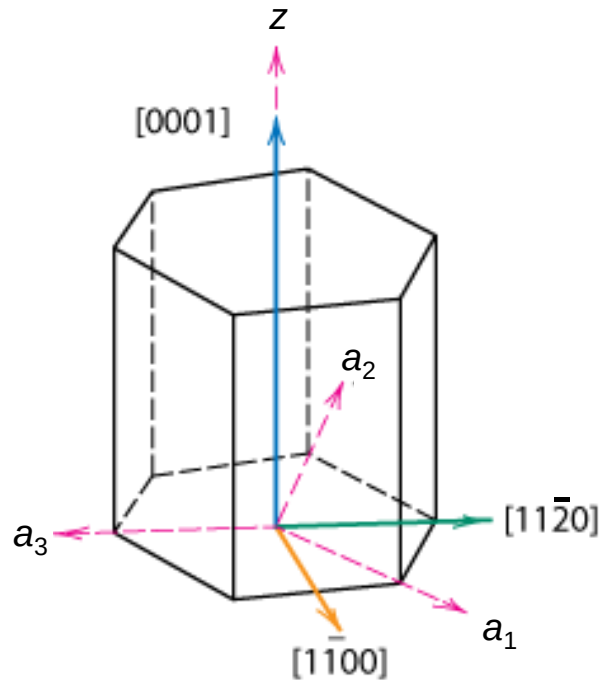


Fig. 3.8(a), Callister & Rethwisch 8e.

$$[u'v'w'] \rightarrow [uvw]$$

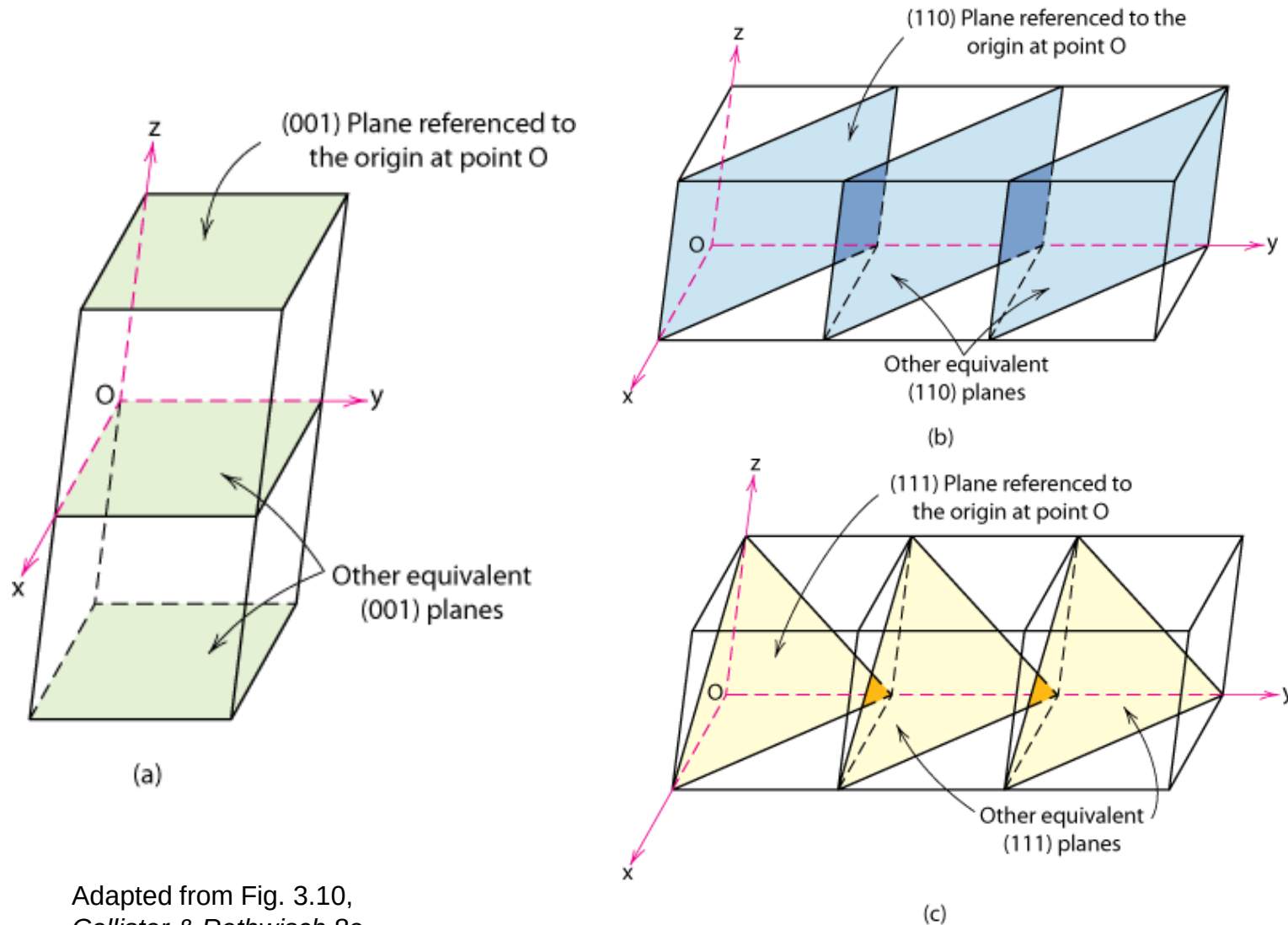
$$u = \frac{1}{3}(2u' - v')$$

$$v = \frac{1}{3}(2v' - u')$$

$$t = -(u + v)$$

$$w = w'$$

Crystallographic Planes



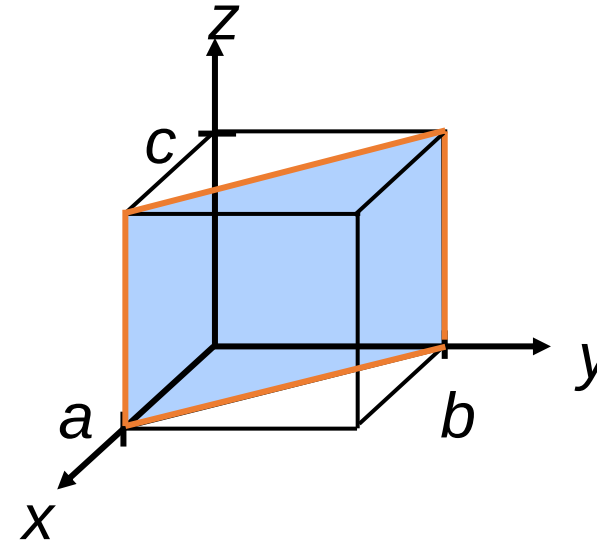
Adapted from Fig. 3.10,
Callister & Rethwisch 8e.

Crystallographic Planes

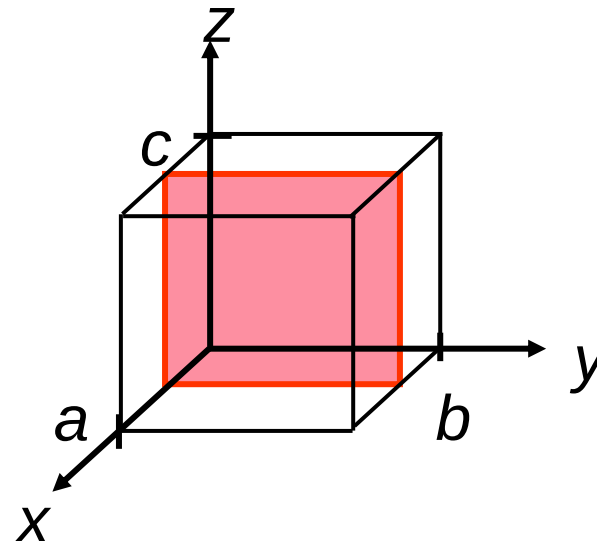
- Miller Indices: Reciprocals of the (three) axial intercepts for a plane, cleared of fractions & common multiples. All parallel planes have same Miller indices.
- Algorithm
 1. Read off intercepts of plane with axes in terms of a, b, c
 2. Take reciprocals of intercepts
 3. Reduce to smallest integer values
 4. Enclose in parentheses, no commas i.e., (hkl)

Crystallographic Planes

<u>example</u>	a	b	c
1. Intercepts	1	1	∞
2. Reciprocals	$1/1$	$1/1$	$1/\infty$
	1	1	0
3. Reduction	1	1	0
4. Miller Indices	(110)		

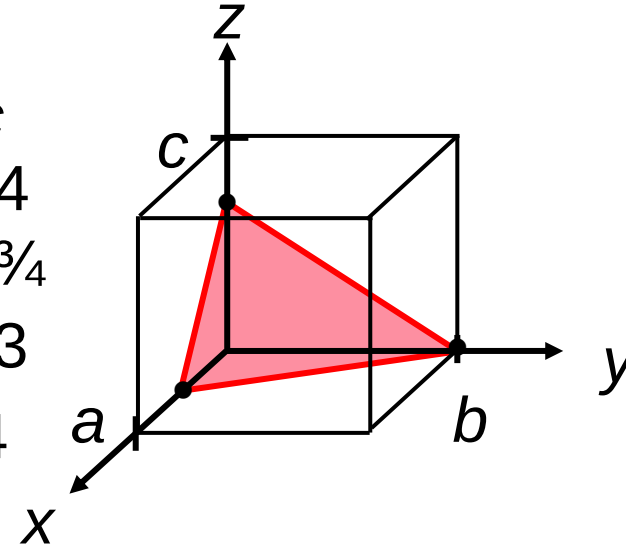


<u>example</u>	a	b	c
1. Intercepts	$1/2$	∞	∞
2. Reciprocals	$1/1/2$	$1/\infty$	$1/\infty$
	2	0	0
3. Reduction	2	0	0
4. Miller Indices	(100)		



Crystallographic Planes

	<u>example</u>	a	b	c
1.	Intercepts	$1/2$	1	$3/4$
2.	Reciprocals	$1/1/2$	$1/1$	$1/3/4$
		2	1	$4/3$
3.	Reduction	6	3	4
4.	Miller Indices	(634)		



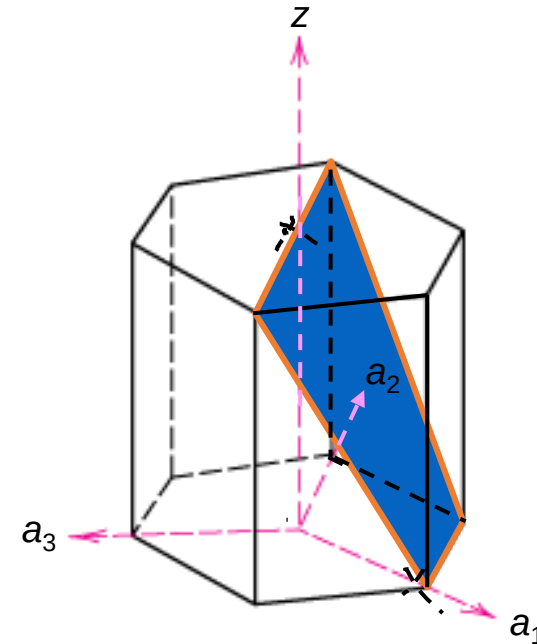
Family of Planes $\{hkl\}$

Ex: $\{100\} = (100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1})$

Crystallographic Planes (HCP)

- In hexagonal unit cells the same idea is used

<u>example</u>	a_1	a_2	a_3	c
1. Intercepts	1	∞	-1	1
2. Reciprocals	1	$1/\infty$	-1	1
	1	0	-1	1
3. Reduction	1	0	-1	1
4. Miller-Bravais Indices	$(10\bar{1}1)$			



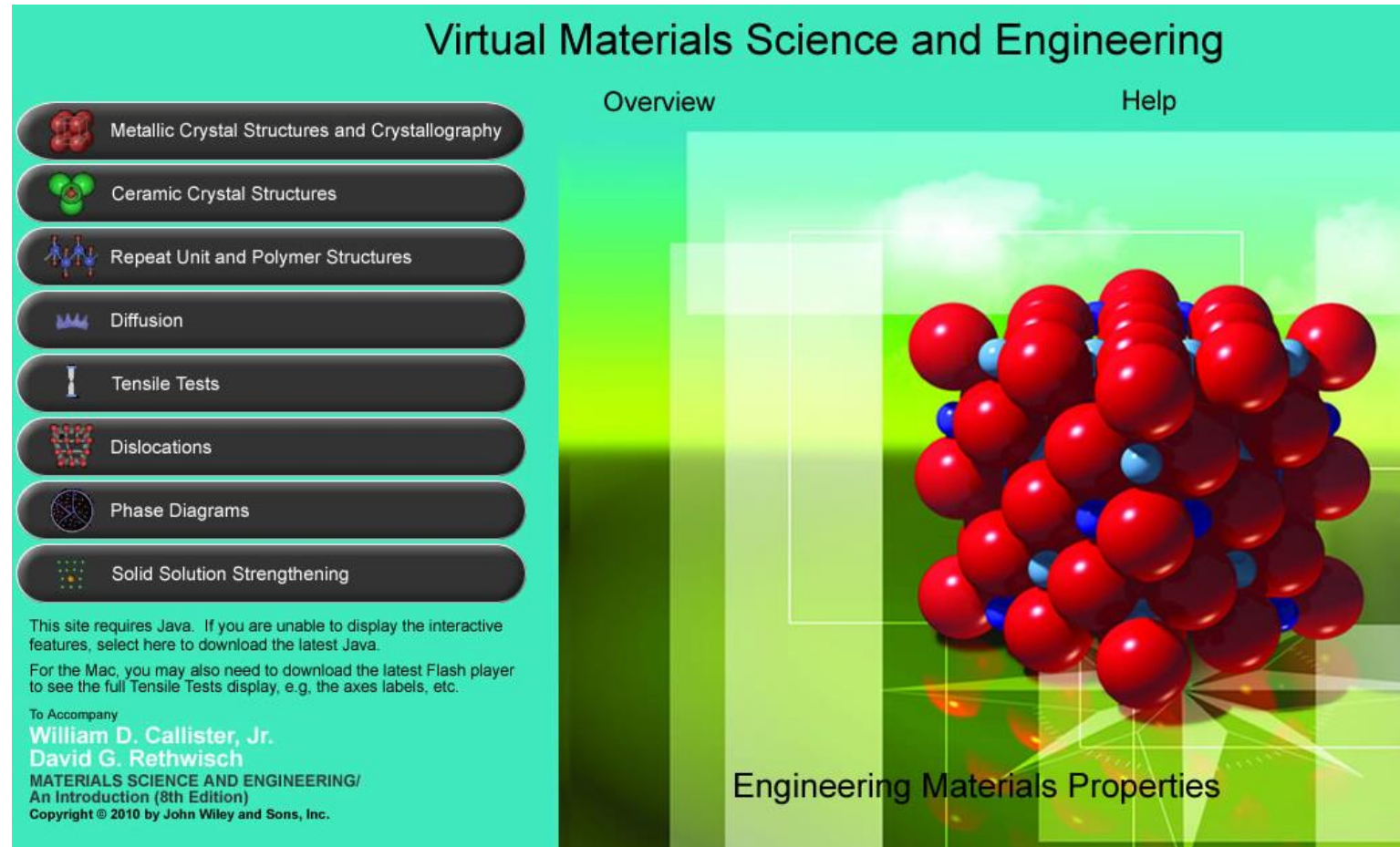
Adapted from Fig. 3.8(b),
Callister & Rethwisch 8e.

Crystallographic Planes

- We want to examine the atomic packing of crystallographic planes
- Iron foil can be used as a catalyst. The atomic packing of the exposed planes is important.
 - a) Draw (100) and (111) crystallographic planes for Fe.
 - b) Calculate the planar density for each of these planes.

Virtual Materials Science & Engineering (VMSE)

- VMSE is a tool to visualize materials science topics such as crystallography and polymer structures in three dimensions



- Available in Student Companion Site at www.wiley.com/college/callister and in WileyPLUS

VMSE: Metallic Crystal Structures & Crystallography Module

- VMSE allows you to view crystal structures, directions, planes, etc. and manipulate them in three dimensions

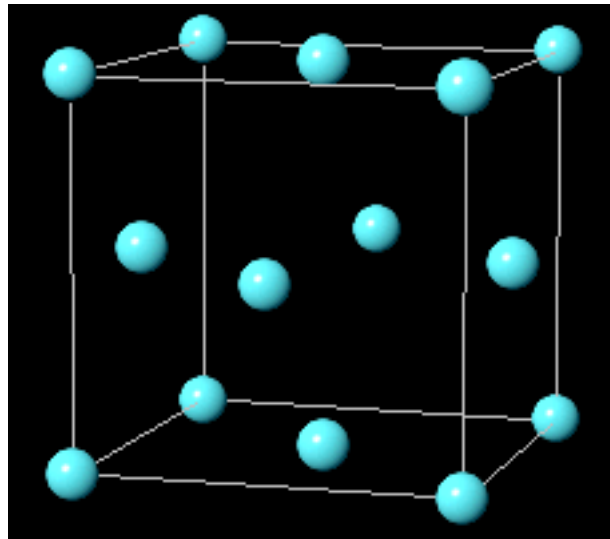
[Main Menu](#) **Metallic Crystal Structures and Crystallography** [Help](#)

In the Crystal Systems and Unit Cells for Metals submodule you may observe unit cells for the seven crystal systems as well as unit cells for face-centered cubic, body-centered cubic, and hexagonal close-packed crystal structures. Rotation of the unit cells is possible (using mouse click-and-drag) in order for you to better visualize them from a three-dimensional perspective. You may also generate crystal structures for FCC and HCP by the stacking of close-packed planes of atoms on top of one another [Close-packed Structures (Metals) submodule]. The Crystallographic Directions and Crystallographic Planes submodules allow for the generation and rotation of crystallography planes and directions. In the Planar Atomic Arrangements—FCC and BCC submodule, atomic arrangements of several crystallographic planes for FCC and BCC may be displayed; these planes may also be rotated. Also, the Molecular Definition Utility allows the user to generate his/her own unit cells and molecular structures.

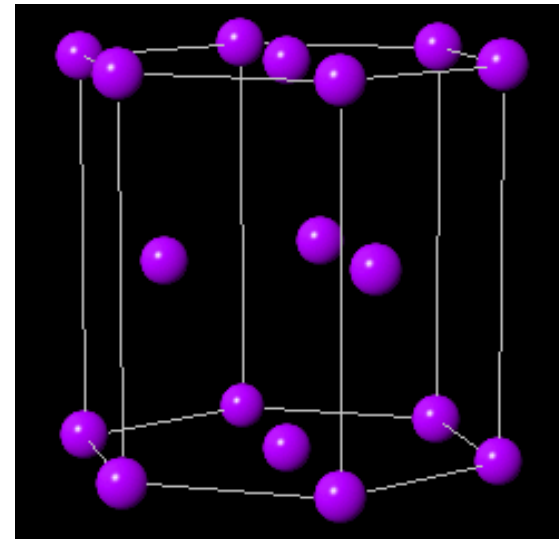
	<u>Crystal Systems and Unit Cells for Metals</u>		<u>Crystallographic Planes</u>
	<u>Close-packed Structures (Metals)</u>		<u>Planar Atomic Arrangements - FCC and BCC</u>
	<u>Crystallographic Directions</u>		<u>Molecular Definition Utility</u>

Unit Cells for Metals

- VMSE allows you to view the unit cells and manipulate them in three dimensions
- Below are examples of actual VMSE screen shots



FCC Structure



HCP Structure

VMSE: Crystallographic Planes Exercises

Main Menu

Module Menu

Crystallographic Planes

Help

This submodule allows you to do the following: (1) Display and rotate (using mouse click-and-drag) unit cells containing the five crystallographic planes listed on the left bar window; (2) Display and rotate unit cells and planes represented in Problems 3.40, 3.41, 3.42 and 3.49 in the text; (3) Test your ability to determine the indices of a given plane ("Exercises"), and (4) Choose a set of indices, and have displayed the

 (0 1 1)

 (1 1 1)

 (2 0 1)

 (1 3 0)

 (3 2 1)

 Prob. 3.40

 Prob. 3.41

 Prob. 3.42

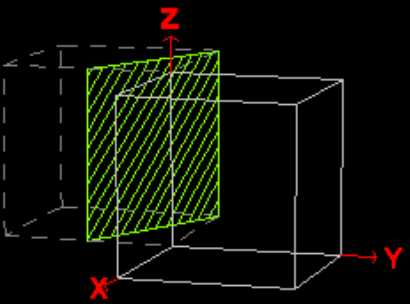
 Prob. 3.49

 Exercises

 Indices

Select the correct set of indices for the plane shown.

(3 $\bar{1}$ 2)
(0 0 $\bar{1}$)
($\bar{1}$ 1 2)
($\bar{1}$ 3 2)
($\bar{1}$ 2 0)
(0 $\bar{1}$ 1)

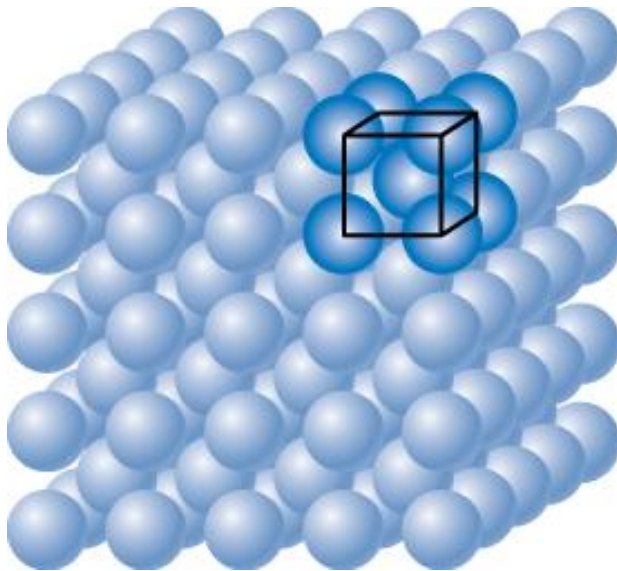


Additional practice on indexing crystallographic planes

51

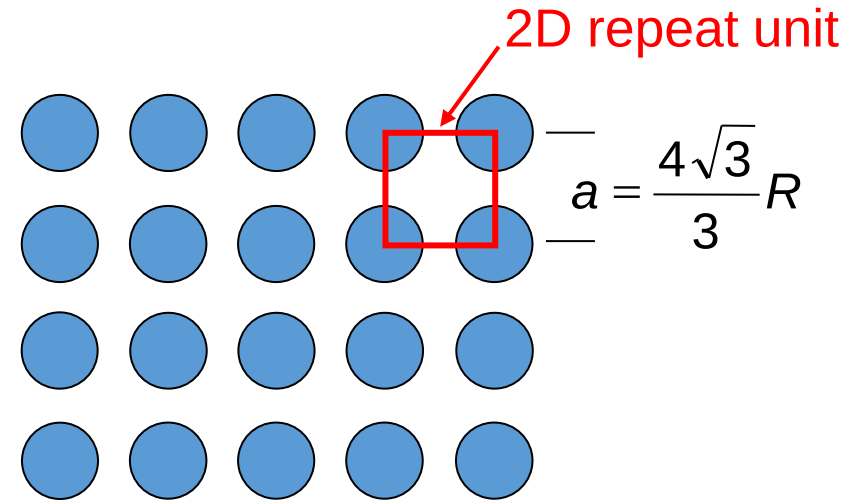
Planar Density of (100) Iron

Solution: At $T < 912^\circ\text{C}$ iron has the BCC structure.



Adapted from Fig. 3.2(c), Callister & Rethwisch 8e.

(100)



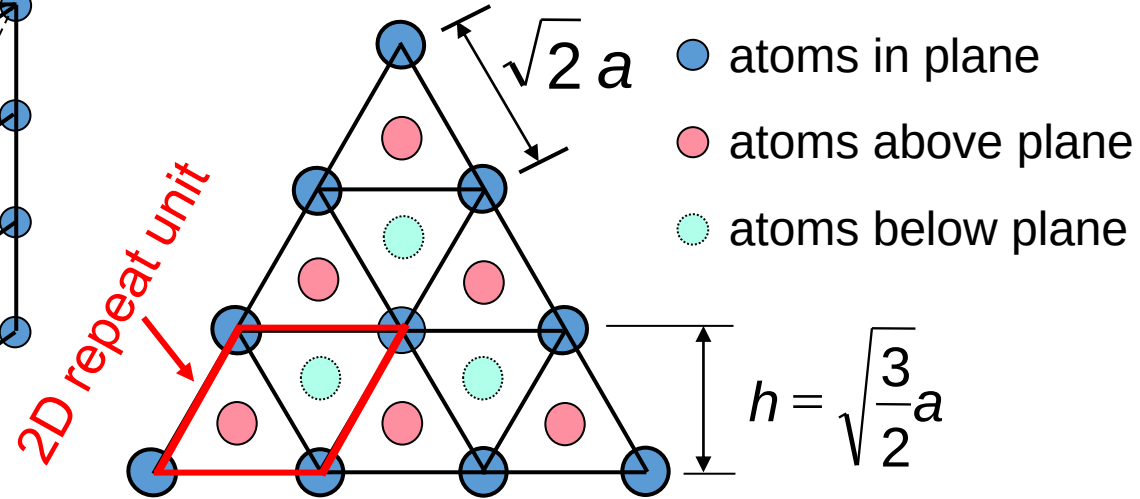
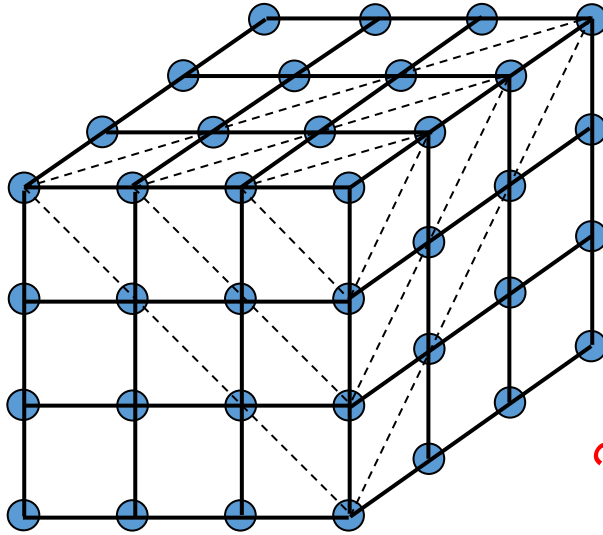
Radius of iron $R = 0.1241 \text{ nm}$

$$\text{Planar Density} = \frac{\frac{\text{atoms}}{\text{2D repeat unit}}}{\frac{\text{area}}{\text{2D repeat unit}}} = \frac{1}{\left(\frac{4\sqrt{3}}{3}R\right)^2} = 12.1 \frac{\text{atoms}}{\text{nm}^2} = 1.2 \times 10^{19} \frac{\text{atoms}}{\text{m}^2}$$

Planar Density of (111) Iron

Solution (cont): (111) plane

1 atom in plane/ unit surface cell



$$\text{area} = \sqrt{2} ah = \sqrt{3} a^2 = \sqrt{3} \left(\frac{4\sqrt{3}}{3} R \right)^2 = \frac{16\sqrt{3}}{3} R^2$$

atoms
2D repeat unit

Planar Density =

area
2D repeat unit

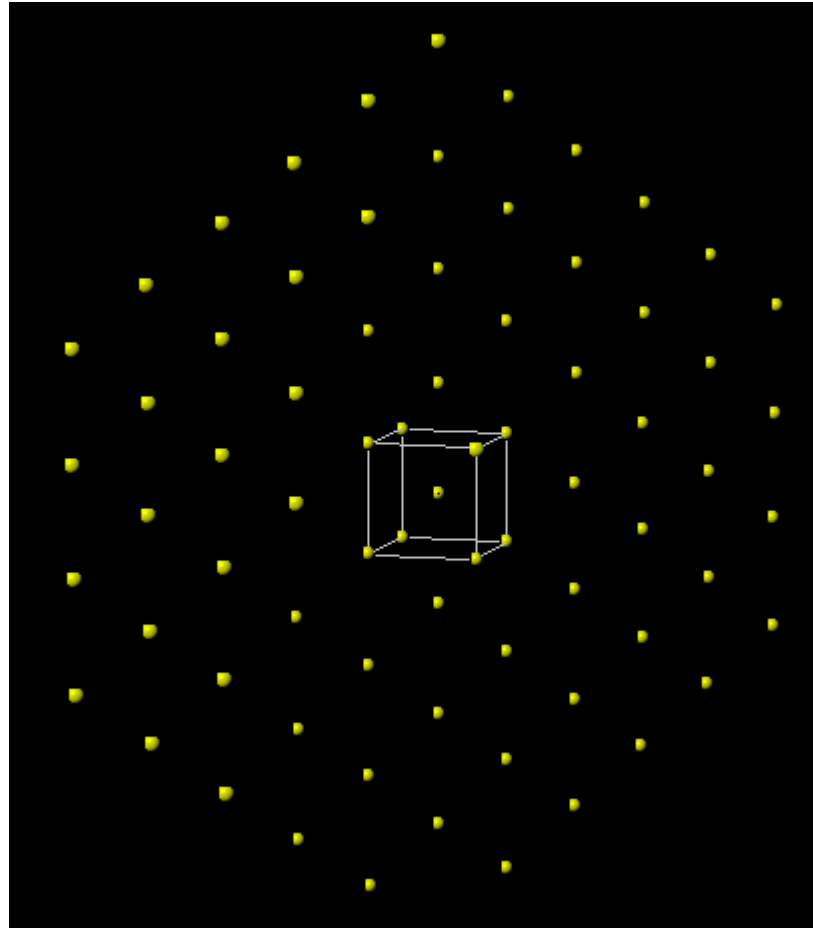
$$= \frac{1}{\frac{16\sqrt{3}}{3} R^2} = 7.0 \frac{\text{atoms}}{\text{nm}^2}$$

$$0.70 \times 10^{19} \frac{\text{atoms}}{\text{m}^2}$$

Planar Density of (111) Cu

VMSE Planar Atomic Arrangements

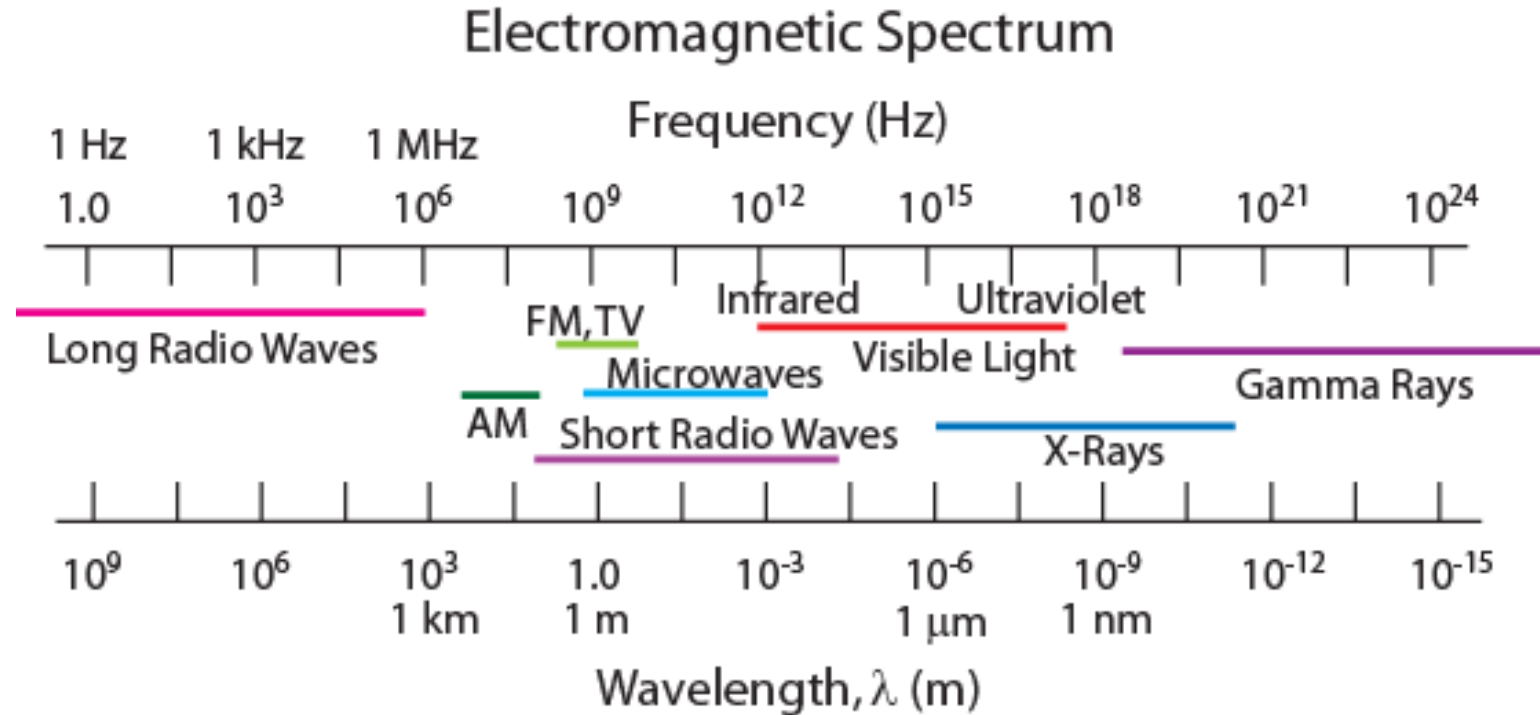
- VMSE allows you to view planar arrangements and rotate them in 3 dimensions



BCC (110) Plane

Planar Density of (110) Iron

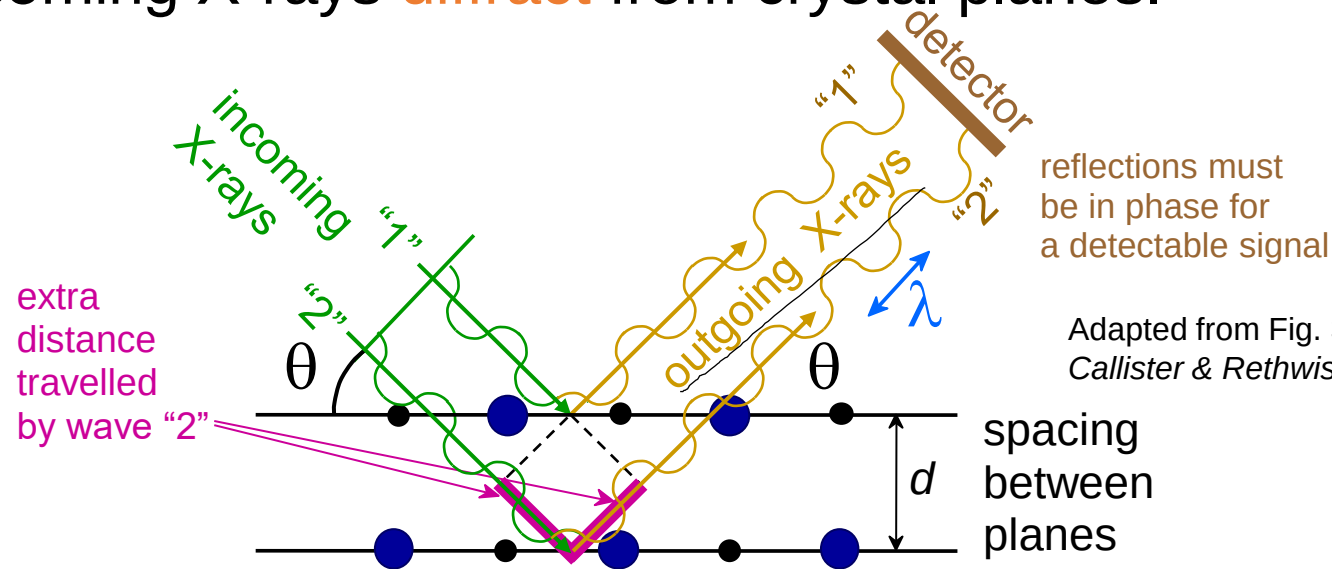
X-Ray Diffraction



- Diffraction gratings must have spacings comparable to the wavelength of diffracted radiation.
- Can't resolve spacings $< \lambda$
- Spacing is the distance between parallel planes of atoms.

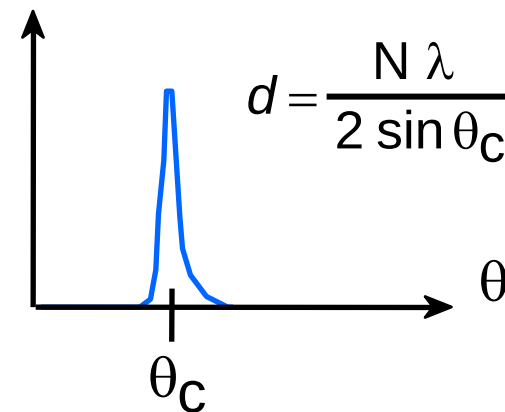
X-Rays to Determine Crystal Structure

- Incoming X-rays **diffract** from crystal planes.

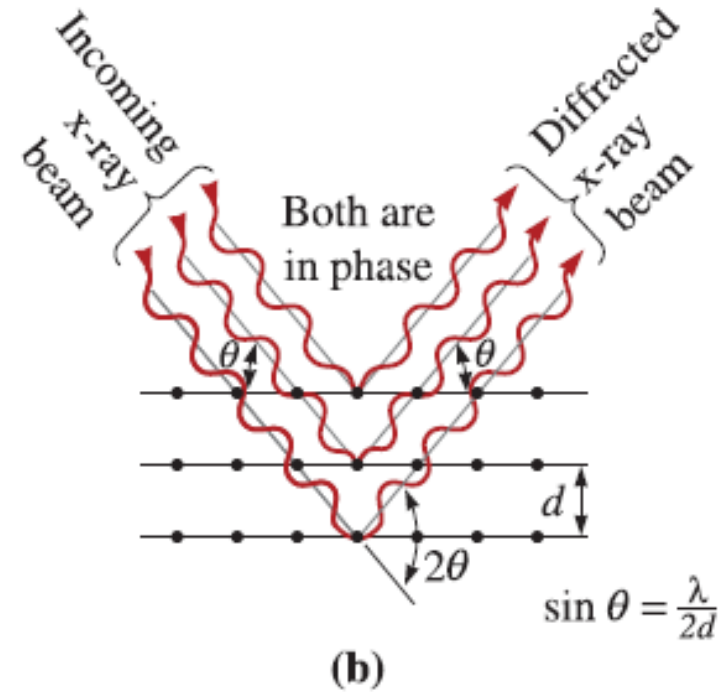
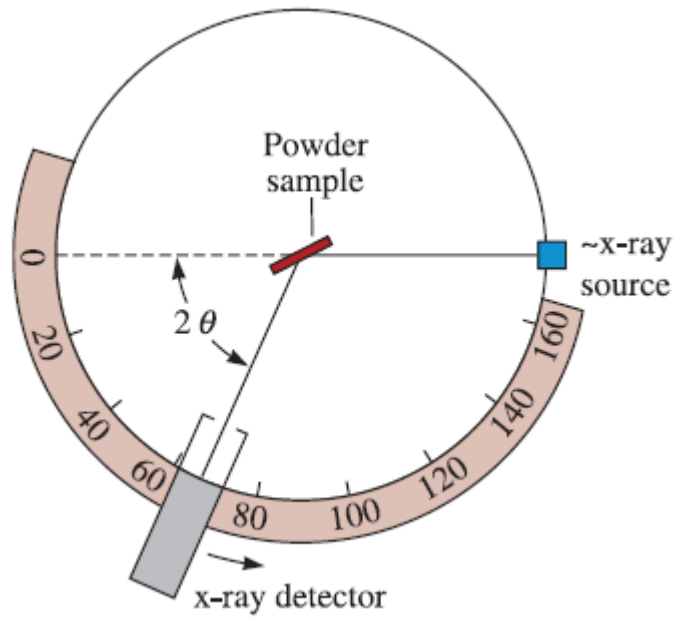


Measurement of critical angle, θ_c , allows computation of planar spacing, d .

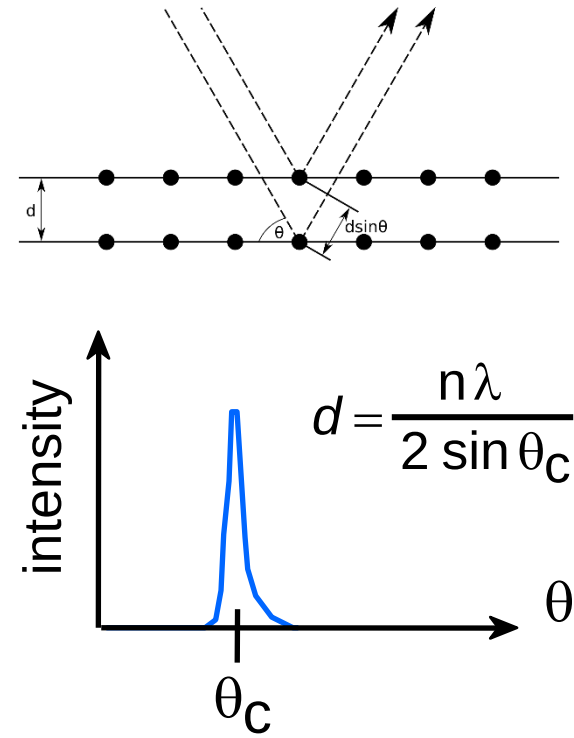
X-ray intensity (from detector)



X-Rays to Determine Crystal Structure



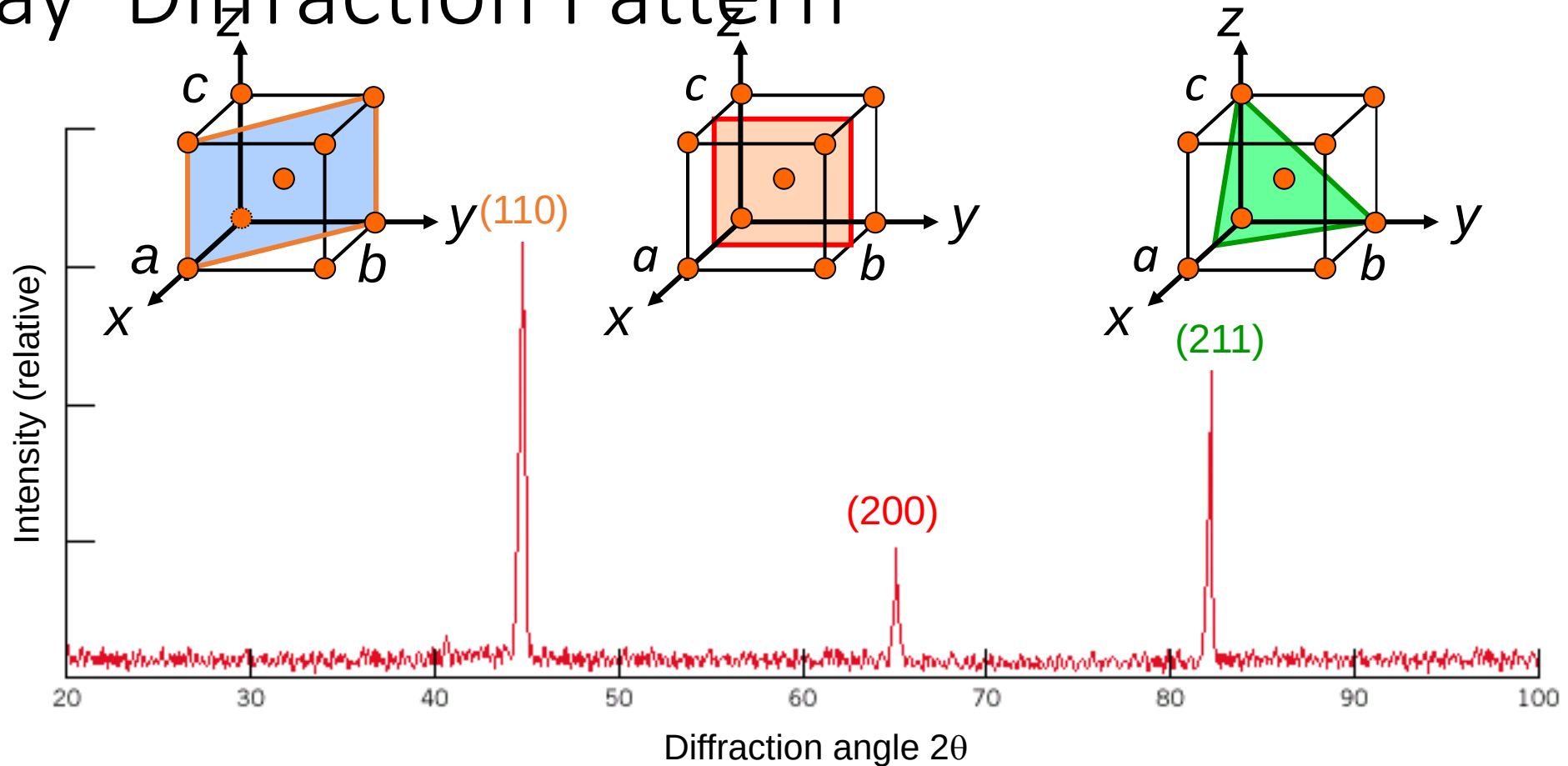
Bragg's Law: Proof



Selection rules

Bravais lattices	Example compounds	Allowed reflections	Forbidden reflections
Simple cubic	Po	Any h, k, ℓ	None
Body-centered cubic	Fe, W, Ta, Cr	$h + k + \ell = \text{even}$	$h + k + \ell = \text{odd}$
Face-centered cubic (FCC)	Cu, Al, Ni, NaCl, LiH, PbS	h, k, ℓ all odd or all even	h, k, ℓ mixed odd and even
Diamond FCC	Si, Ge	All odd, or all even with $h + k + \ell = 4n$	h, k, ℓ mixed odd and even, or all even with $h + k + \ell \neq 4n$

X-Ray Diffraction Pattern



Diffraction pattern for polycrystalline α -iron (BCC)

Adapted from Fig. 3.22, *Callister 8e*.

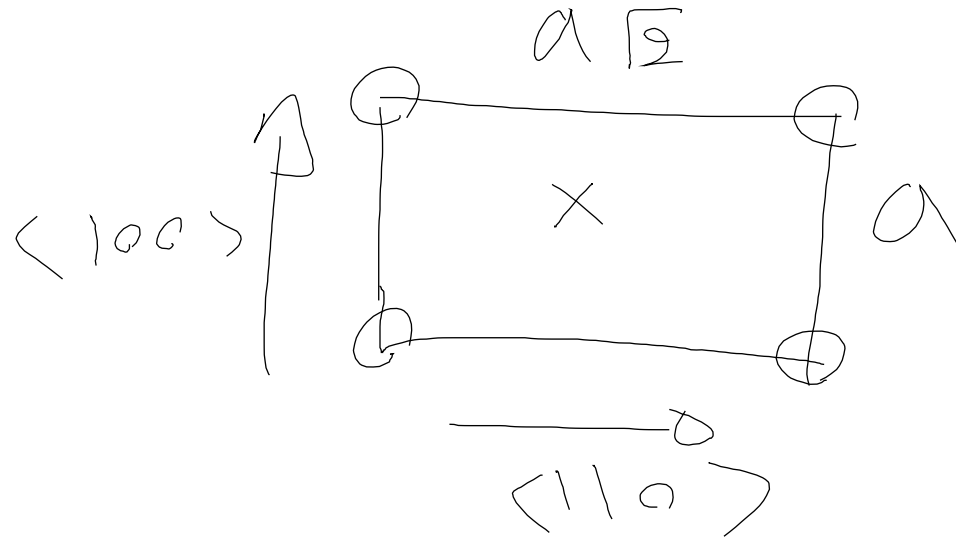
SUMMARY

- Atoms may assemble into **crystalline** or **amorphous** structures.
- Common metallic crystal structures are **FCC**, **BCC**, and **HCP**. **Coordination number** and **atomic packing factor** are the same for both FCC and HCP crystal structures.
- We can predict the **density** of a material, provided we know the **atomic weight**, **atomic radius**, and **crystal geometry** (e.g., FCC, BCC, HCP).
- **Crystallographic points**, **directions** and **planes** are specified in terms of indexing schemes.
Crystallographic directions and planes are related to **atomic linear densities** and **planar densities**.

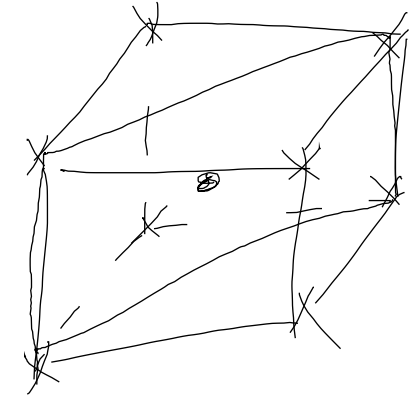
SUMMARY

- Materials can be **single crystals** or **polycrystalline**.
Material properties generally vary with single crystal orientation (i.e., they are **anisotropic**), but are generally non-directional (i.e., they are **isotropic**) in polycrystals with randomly oriented grains.
- Some materials can have more than one crystal structure. This is referred to as **polymorphism** (or **allotropy**).
- **X-ray diffraction** is used for crystal structure and **interplanar spacing** determinations.

Planar Density of (110) Iron



$$\text{Density}_2 = \frac{4 \times \frac{1}{4} + 1}{a \times a\sqrt{2}} = \frac{2}{\sqrt{2} a^2} = \frac{\sqrt{2}}{16R^2}$$



$$a\sqrt{3} = 4R$$

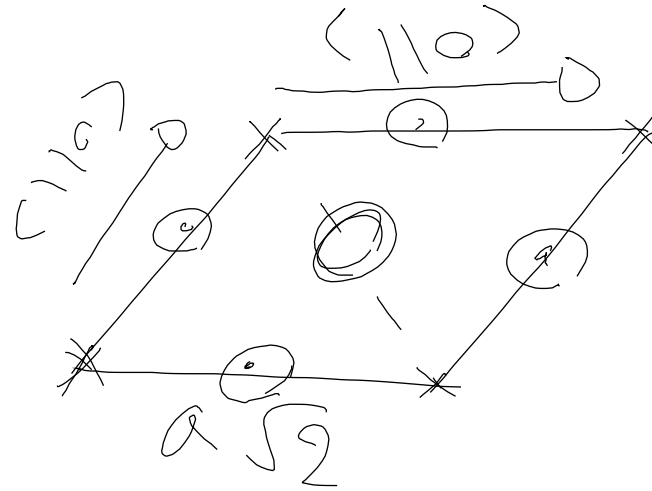
$$a = \frac{4}{\sqrt{3}} R$$

$$a^2 = \frac{16}{3} R^2$$

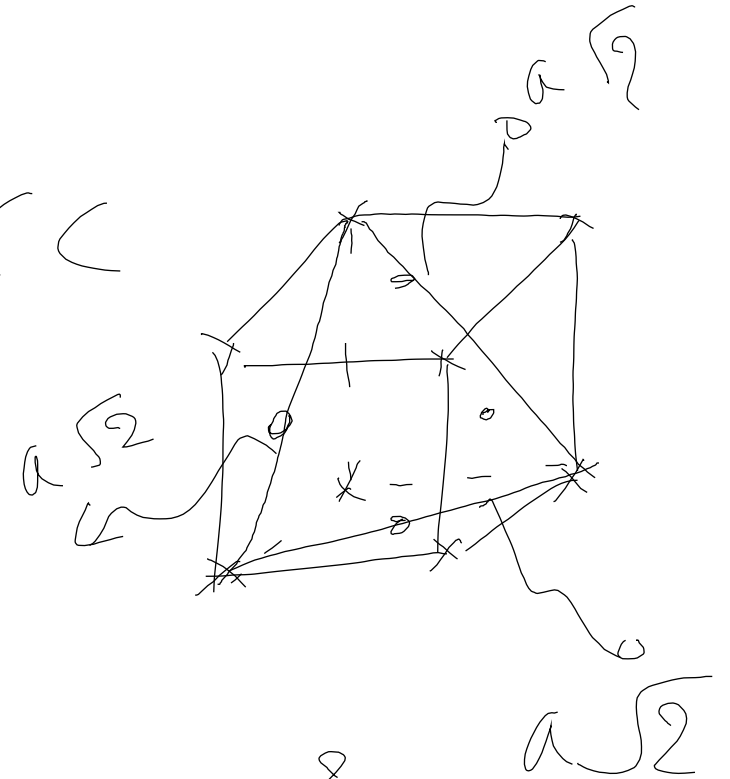
Planar Density of (111) Cu

$$a\sqrt{2} = 4R$$

$$a = 2\sqrt{2}R$$



FCC

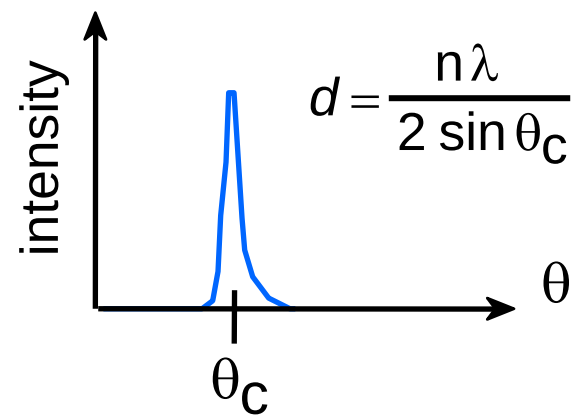
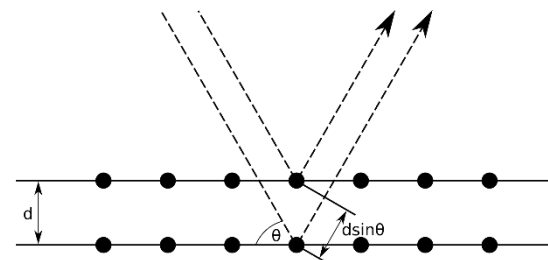
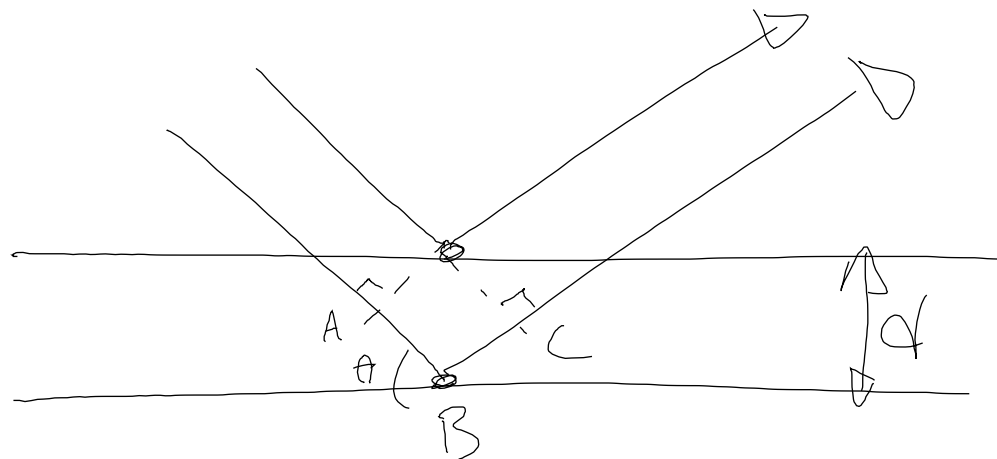


$$\text{Area} = \sqrt{3} a^2 = 8\sqrt{3} R^2$$

$$\text{Density} = \frac{1 + 4 \times \frac{1}{2} + 1}{8\sqrt{3} R^2} = \frac{1}{2\sqrt{3} R^2} = \frac{\sqrt{3}}{6 R^2}$$

Bragg's law: proof

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad (h, k, l)$$



$$\text{Path diff.} = \overline{AB} + \overline{BC}$$

$$\overline{AB} + \overline{BC} = n\lambda$$

$$\overline{AB} = \overline{BC} = d \sin(\theta) \quad \therefore 2d \sin(\theta) = n\lambda$$